Product Relatedness in Tasks: A Novel Measure and Multi-Product Firms

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Abstract

Products vary in the tasks they require and, consequently, in the occupations they draw on. I develop a novel measure of product relatedness grounded in the occupational composition of single product manufacturers in Denmark. Using matched employer–employee data, I represent each product as a vector of task requirements and construct a bilateral product cosine similarity measure, weighted by occupational flows. To take the measure to multi-product firms, I anchor each firm to its core product and compute the task distance between that core and every candidate product, linking these distances to the firm's product-scope decisions. Multi-product firms expand into products closely related to their core and drop products that are more distant in task space, consistent with rising marginal costs of producing away from core competences. These findings highlight the role of workforce composition in shaping firm scope and offer a task-based perspective on product diversification.

Keywords: Multi-Product Firms, Product Scope, Core Competences, Task Similarity, Product Relatedness

JEL codes: L23, L25, J24

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1 Introduction

Previous literature has shown that multi-product firms (MPFs) account for a disproportionately large share of production, exports, and employment (Irlacher (2024) provides an overview), yet we still know relatively little about how they make decisions on *which* products to add (drop) in the product mix from an empirical perspective. This paper studies product diversification (retraction) from the perspective of a firm's labor-based core competences. The central idea is simple: products differ in the occupations and tasks required to produce them, and firms should find it easier to expand into products that draw on capabilities already present in their workforce.

This perspective is closely related to classic resource-based views (RBV) of the firm, which place the extensive margin of production as a key driver of growth. Specifically, in Penrose (1959), firm growth reflects the accumulation and redeployment of internal resources in new product lines, while Schumpeter (1951) emphasizes innovation and product diversification through the recombination of existing capabilities coined "creative destruction". The RBV literature further highlights flexibility in manufacturing and dynamic capabilities as sources of competitive advantage, especially in times of increasing globalization and rapid technological change (Teece et al., 1997). Applied to MPFs, these ideas imply that product diversification is shaped not only by demand conditions, but also by whether firms can reuse and reconfigure what they already know how to do. A central theme in the MPF literature in this context is that firms actively adjust scope around a set of core products or core competences (as in Eckel and Neary (2010) for instance). The question, then, is how to measure those underlying capabilities in a way that is observable in the data.

In this paper, I propose to do so using occupational structures. Occupations provide a tractable empirical proxy for the tasks required for production. If two products rely on similar occupational compositions, they are likely to require similar tasks and capabilities; if their occupational structures differ substantially, moving between them should be more costly. Accordingly, I argue that product-adding decisions depend positively, and product-dropping decisions depend negatively, on occupational similarity.

Aggregate data seems to support the idea of firms utilizing incumbent resources to grow:

Firms are generally expanding in worker count within existing occupations. Between 2010 and 2019, the average Danish manufacturing firm increased the number of products it manufactured by roughly 12 percent and its number of employees by roughly 22 percent, yet the number of distinct occupations it employed (also referred to as 'organizational breadth') grew by only 5 percent. Figure 1 illustrates the growth of the average number of employees, products, and occupations, indexed in 2010.

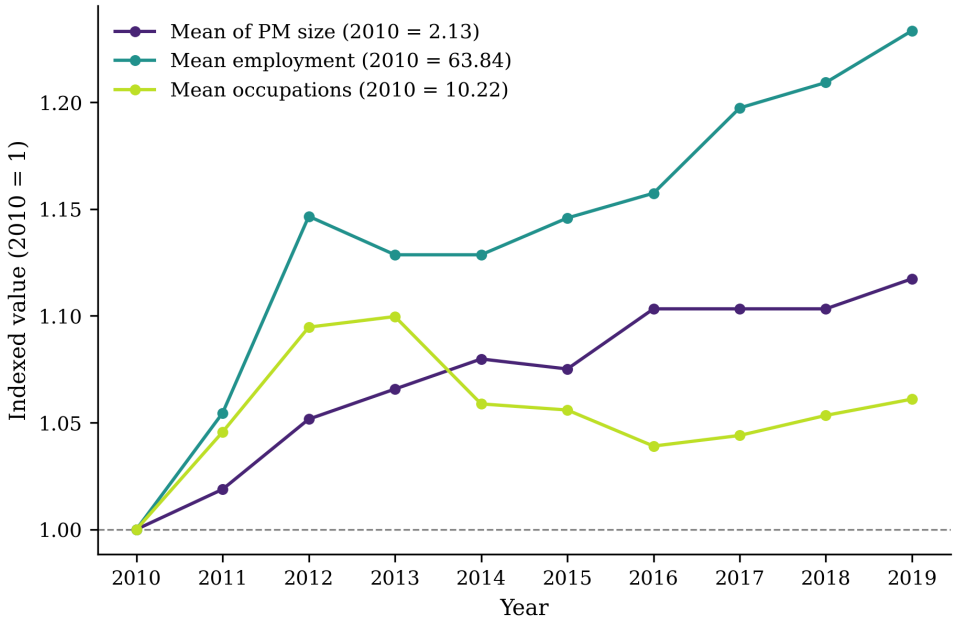


Figure 1: *This figure displays the average number of products, employees, and unique occupations in a sample firm across years. Products are grouped at the four-digit level, and occupations at the three-digit level. Data come from a matched employer-employee dataset covering Danish manufacturing firms.*

Section 2 describes the data used to analyze this link between products and occupations. The sample consists of a matched employer-employee dataset covering all Danish manufacturing firms and their employees from 2010 to 2019. Key variables include product-level sales at the firm-year level, and occupational DISCO codes linked to the associated employees. I then proceed to present three novel stylized facts endogenous product choices and the link between firm scope and organizational breadth. First, product compositions are not random, but rather, there is great heterogeneity in the likelihood of two product pairs being co-produced, and that the majority of economic activity comes from firms producing across multiple broad product categories. Secondly, I confirm the indications of Figure 1 at the

firm-level: larger scope correlates with more workers but not more occupations. Lastly, I consider the connection between organizational breadth and concentration with firm scope (under construction).

Section 3 constructs the relatedness measure. For each product, I compute the share of labor hours accounted for by each occupation among the single product firms (SPFs) that produce it, yielding a product-level vector of occupational intensities (similar to Boehm et al. (2022)). Bilateral product similarity is calculated as the cosine similarity between two such vectors, extended with a weighting matrix that increases the similarity if two products require occupations that are not identical but share tasks (two adjacent types of engineers, for instance). The weights are estimated from excess worker flows between occupation pairs, following Neffke and Henning (2013) and, at the occupation level, Joyez and Laffineur (2022). The resulting similarity matrix organizes the product space into clusters (dominant clusters include food products, plastics and rubber, and a large machinery cluster) that regularly cross aggregate product chapters. In a robustness check, I show that the hour-weighting changes the similarity structure substantially compared to similarities of extensive margin presence, and that the weighting matrix influences ordinal similarity structures more than cardinal. The occupation-based product similarity structure differs greatly from a corresponding similarity structure based on intermediate inputs.

Section 4 embeds this measure in Eckel and Neary's framework by letting the cost penalty of adopting a candidate product be governed by the task distance to the firm's core product. This turns the competence ladder into an object that is measured, and yields testable monotonicity predictions: the probability that a firm adds a candidate product is increasing in task similarity, and the probability that it drops an incumbent product is decreasing in it.

Section 5 takes the monotonicity predictions of Section 4 to the data, considering only MPFs not included so far, and finds support for both margins. Portfolios are organized in task space exactly as the model predicts: mean similarity to a firm's core product declines monotonically with a product's rank in firm sales. On the extensive margin, I estimate that task similarity to a firm's core product significantly raises the probability that the firm adds a given candidate, by more than five times the unconditional probability that any

candidate is added in a given year. This relationship is found to be convex, with the increase concentrated among the top decile of most similar candidates. The dropping side mirrors this pattern: similarity significantly lowers the probability that an incumbent product is dropped, an effect concentrated among the least similar, lowest-ranked products in a firm's portfolio. Both results are robust to replacing the core product with portfolio-wide similarity aggregators and to holding the candidate pool fixed across firms and years.

Section 6 concludes.

Related literature. This analysis contributes to two main strands of literature.

Firstly, this paper speaks to an extensive literature on MPFs, which documents their importance for aggregate revenue, exports, and employment and asks how they choose what to produce.[1] Bernard et al. (2005; 2010; 2011) document that product scope and coproduction patterns are far from random, establishing product adding and dropping as an important margin of firm growth. Eckel and Neary (2010) model flexible manufacturing[2] in which competition and demand shocks push MPFs to concentrate on their core competence. Nocke and Yeaple (2008; 2014) show how heterogeneous firm capabilities generate a trade-off between scope and marginal cost that shapes the size distribution of MPFs and their response to globalization. Mayer et al. (2014) show that tougher destination-market competition induces firms to skew sales toward their best-performing products. Arkolakis et al. (2021) estimate that products farther from a firm's core competence carry higher production costs even as market-access costs exhibit economies of scope. Macedoni and Xu (2022) document that firms differ not only in productivity but in their flexibility to introduce new varieties cheaply. Boehm et al. (2022) provide evidence that shared intermediate input capabilities drive firms' industry choices. This paper directly relates to this by showing that diversification choices are guided by task-overlap between the core competence of the firm and the

[1] See also Bernard et al. (2007) for a summary of firm-level trade facts in relation to product scope; Iacovone and Javorcik (2010), Eckel et al. (2015), Eckel and Irlacher (2017), and Eckel and Yeaple (2024) on further margins of multi-product exporting, quality choice, offshoring, and labor-market sorting; Alvarez et al. (2016) on the productivity consequences of product-mix changes; Goldberg et al. (2009; 2010) on MPFs and imported inputs; Mayer et al. (2021) and Koh and Raval (2025) on further margins of product mix and shared-input economies of scope; and Irlacher (2024) for a recent survey.

[2] Eckel and Neary define 'flexible manufacturing' as *"the ability of firms to add varieties with a minimum of adaptation"*. This paper will use the same phrase accordingly.

requirements for a candidate product. Hence, this paper provides a new empirical approach to assessing firms' latent marginal costs relative to its core competence.

Secondly, this paper contributes to the literature on relatedness and diversification, which asks how the similarity of economic activities disciplines the direction of firm, regional, or national growth. Lemelin (1982) estimates the probability of diversifying into a candidate industry conditional on its relatedness to the firm's core product within a Penrose-Rubin framework (Rubin, 1973). Chatterjee and Wernerfelt (1991) link the type of resources a firm holds to the type of diversification it undertakes; either related or unrelated. Hidalgo et al. (2007) introduce the "product space," mapping relatedness between products through their propensity to be co-exported by the same countries.[3] Some papers also study shared underlying resources directly, building on RBV. Farjoun (1994) proposes so-called resource-based industry groups, identified by clustering firms' lines of business by occupational profile, and shows that diversification concentrates within these groups. Neffke and Henning (2013) infer skill relatedness between industries directly from excess worker flows and show that firms diversify preferentially into industries linked by this measure, more so than into industries linked by value chains or by shared classification codes. Also here, Boehm et al. (2022) contribute by providing a measure of product relatedness based on cosine similarities of single-industry *plants*' expenditure shares across all intermediate inputs.[4] This paper applies the same idea of using single-*product firms*' (not industry and plants) occupational inputs to proxy for task inputs. This furthermore allows me to include a weighting matrix in the similarity measure, which accounts for partial overlap in tasks between occupations; both novel in the existing literature.

[3]The product-space framework is extended by Hausmann et al. (2007), and Hausmann and Hidalgo (2011), and applied to firm-level adoption decisions by Macedoni et al. (2024).

[4]Neffke et al. (2011) extend the skill-relatedness logic to regional diversification, and Joyez and Laffineur (2022) apply the excess-flow methodology at the occupation level, constructing an occupation space for the French labor market. This approach builds on a broader labor literature establishing that occupations carry portable, task-specific human capital, including Autor et al. (2003), Gathmann and Schönberg (2010), Poletaev and Robinson (2008), Ingram and Neumann (2006), and Goos and Manning (2007).

2 Firms and Products; Workers and Occupations

To establish a credible link between products and occupations, this study leverages registry data covering the universe of Danish manufacturing firms, their yearly product-level sales, and associated worker records including occupational codes. This section describes the sample only as far as is needed to read the results, and then presents three stylized facts on the joint behavior of product scope and occupational structure. These facts motivate the measurement exercise and the subsequent analysis. Full detail on sample construction, variable definitions, and summary statistics is deferred to Appendix A.

2.1 Data and sample

I combine three sources of Danish registry data, linked by a unique firm identifier and year. Firm-product sales come from the VARS database, which records sales at up to ten digits of the Combined Nomenclature for all manufacturing firms with more than ten employees. Worker records, including occupational codes and hours, come from the IDA databases (IDAP, IDAN, and INDH). I restrict the sample to 2010–2019 to avoid contamination from the financial crisis and from Covid-19 mitigation policies, and to avoid breaks in the occupational classification. The resulting sample contains 3,560 manufacturing firms and 370,072 workers.

Two aggregation choices shape everything that follows. Products are measured at the four-digit Harmonized System (HS) level. This retains economically meaningful distinctions: "0203: Meat of swine" is separated from "0204: Meat of sheep and goats", while discarding distinctions that carry no plausible implication for task requirements, such as the six-digit split of swine meat into fresh and frozen.[5] Occupations are measured at the three-digit DISCO level, the Danish implementation of ISCO. Two digits is too coarse for a measure built on bilateral occupational proximity: "21: Work in Science and Engineering" pools activities that a task-based measure must keep apart, and at three digits it resolves into six sub-codes including "214: Engineering Professionals" and "215: Electrotechnology Engineers." Four digits adds detail: for instance, "413: Text and Data Processing" which splits into text

[5]The methodology applies without modification at finer levels of product aggregation.

processing and data registration, but it is unlikely to correspond to distinct product-level capabilities. The sample contains 167 unique three-digit occupations.[6]

Remaining construction choices are conventional and are described in Appendix A.1: outlier trimming on revenue, employment, and product counts; concordance of all HS codes to 2007 correspondences; CPI deflation to 2015 prices; and restriction of the worker sample to ages 25–60 with the highest DISCO quality flag. Appendices A.2 and A.3 report firm-, product-, worker-, and occupation-level summary statistics. Four magnitudes are worth bringing: the average firm employs 84.6 workers across 10.9 occupations, more than half of all firms produce a single product across the entire sample period, while the sample average number of products is 2.3.

2.2 Product mix changes, occupational adding and dropping

When a firm expands the number of products it manufactures, it is likely to increase its demand for labor inputs. The *type* of that increase, however, depends on the nature of the product(s) adopted; a highly endogenous firm-decision. This subsection presents three stylized facts on products, occupations, and the link between these.

Fact 1 establishes that co-production is systematic rather than random, and the most common product pairings frequently cross two-digit product boundaries. **Fact 2** shows that firms seem to target their increased labor demand towards already incumbent labor types: broader product scope is associated with substantially more workers but not with more occupations, conditional on scale (revenue). This holds both across firms within the same sector-year cell and within the firm. **Fact 3** shows that a shared core product predicts *how* firms allocate hours across occupations far more strongly than *which* occupations they employ.

[6] 2.52 percent of workers register a third DISCO character as 0 in at least one sample year, and are thus not encompassed by the three-digit civilian codes. Workers with a non-civilian code registered in any year account for a comparable 3.14 percent of total wages and 3.59 percent of total hours. I keep these in the sample for two reasons: 1) they represent meaningful task inputs (requirement for DISCO quality is high), and 2) I handle task-overlap explicitly via estimation of bilateral occupational excess worker flows (cf. Section 3.2), which makes sure no occupation is left isolated in the occupation space. The latter point is especially relevant, as restricting to workers who are *only ever* observed in non-civilian occupations, just 2,043 workers — 0.55 percent of the sample — qualify; the wage and hours bills of this group are smaller still, at a mere 0.17 and 0.20 percent respectively. See Table A1 in Appendix A.1 for the full breakdown.

Fact 1: Co-production is systematic and crosses classification boundaries. Products differ sharply in how often and how intensively they are manufactured, and firms combine them in patterns that are far from random. Table A4 in Appendix A.2 ranks products by the number of manufacturers and by mean sales. The rankings overlap only partially. "7308: Structures of iron or steel" leads on both margins, and "9403: Furniture" and "4418: Wood" appear in both. But "7326: Iron or steel", "3926: Plastics", and "4911: Printed matter" are common in firms' product sets while generating low mean sales, and "2309: Dog or cat food", "9021: Orthopaedic or fracture appliances", and "6810: Cement, concrete or artificial stone" generate high mean sales but are only manufactured by few firms. The first group has the character of support or by-products: for instance, "7204: Ferrous waste and scrap" ranks fourth in number of manufacturers and appears in the two most frequent co-production pairings as is shown in Table C5 in Appendix A.2, yet records low average per-producer sales.

Extending on this idea, Table C5 in Appendix A.2 furthermore shows that some bilateral product pairs are far more likely to be co-produced than others: "7204: Ferrous waste and scrap" appears alongside "7308: Structures of iron or steel" and "7326: Iron or steel" in 81 and 78 firms respectively. Crucially, five of the ten most common pairings reported in Table A5 in Appendix A.2 join products from *different* two-digit categories, including "7326: Iron or steel" with "7616: Aluminium" and "3926: Plastics" with both "8480: Moulds" and "7326: Iron or steel". Others are within-chapter, such as "9401: Seats" with "9403: Furniture" and "4902: Newspapers, journals and periodicals" with "4911: Printed matter".

This boundary-crossing is not confined to a handful of prominent pairs, and it is concentrated among the firms that matter most economically. Table A6 in Appendix A.2 shows that 84 percent of total revenue and 80 percent of employment in the sample comes from firms producing across multiple two-digit HS chapters, although making up just 30 percent of firms. Restricting to firms producing more than one product sharpens the pattern further: 69 percent of this sample produce across HS2 chapters, and these firms generate 94 percent of revenue and 93 percent of employment within the sample. Cross-chapter co-production is therefore close to the norm, not the exception.

Taken together, this is strong indication that the product classification is a poor guide

to production relatedness. If firms combined products according to their position in the HS hierarchy, co-production should concentrate within chapters, and the firms doing most of the co-producing would be a small, unrepresentative slice of the economy. Neither holds. Whatever governs which products appear together inside a firm is therefore only partially captured by classification distance, which motivates constructing a measure of proximity from the production factor firms actually reuse: their workforce.

Fact 2: Targeted hiring within existing occupations. The second fact expands on the simple but revealing insight from Figure 1: the average firm grows in scope throughout the sample period and grows simultaneously in employment, but grows much less in the number of unique occupations it employs.

The aggregate pattern could in principle reflect composition, if firms that happen to expand scope differ systematically from those that do not. I therefore ask whether the same association holds within firms, by regressing the log number of employees and the log number of occupations, respectively, on the log number of products manufactured:

$$\log(y_{jt}) = \beta_1 \log(|\Omega_{jt}|) + \beta_2 \log(R_{jt}) + \lambda_j + \lambda_{s(j)\times t} + \varepsilon_{jt}, \tag{1}$$

where y_{jt} denotes either the number of employees or the number of occupations of firm j in year t, $|\Omega_{jt}|$ denotes product mix size, and R_{jt} denotes firm revenue. Hence β_1 measures the product-scope elasticity of employment and β_2 measures the product-scope elasticity of occupational breadth. No result should be interpreted as causal, as all variables are endogenous to the firm. $s(j)$ denotes the two-digit HS chapter in which firm j records the majority of its sales.

Firm fixed effects absorb firm-specific factors influencing the number of products, including scale, managerial capabilities, time-invariant dependencies in labor demand and technology, and geography. Sector-year fixed effects absorb common shocks to all firms in the same sector: macroeconomic conditions, policy changes, sector-level technological change, and, importantly, common input prices and shared demand trends. The remaining variation captures firm-specific changes that are unusual both within the firm and relative to sector

Table 1: Employment and occupations in scale and scope

	$\log(\text{employees}_{jt})$		$\log(\text{occupations}_{jt})$	
	(1)	(2)	(3)	(4)
$\log(\lvert\Omega_{jt}\rvert)$	0.0882***	0.0517***	0.0377*	0.0234
	(0.0175)	(0.0148)	(0.0164)	(0.0161)
$\log(\text{revenue}_{jt})$		0.4266***		0.1669***
		(0.0146)		(0.0121)
Firm FE	Yes	Yes	Yes	Yes
Sector × year FE	Yes	Yes	Yes	Yes
Observations	22,938	22,938	22,938	22,938
R^2	0.962	0.972	0.891	0.894
Within R^2	0.005	0.258	0.001	0.031

Notes: Clustered (firm) standard errors in parentheses.
$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

peers.

Broader product scope is associated with substantially larger employment. Its association with occupational breadth is weaker throughout, and disappears once scale is held fixed: column (3) shows a marginally significant unconditional association between scope and the number of occupations, which is fully absorbed once revenue enters in column (4). Firms expanding their product portfolios therefore scale primarily by adding workers within a relatively stable set of occupations, rather than by proportionally widening the range of labor types they employ.

The within-R^2 row of Table 1 sharpens this asymmetry. Conditioning on revenue raises the within-R^2 for employment fifty-fold, from 0.5 to 25.8 percent, but leaves the within-R^2 for occupations essentially at its floor, rising from 0.1 to 3.1 percent. In other words, once a firm's change in scale is known, most of its remaining employment growth is accounted for, while its occupational breadth remains almost entirely unexplained by the same scale variation. Employment growth is therefore a close proxy for scale; occupational breadth is not, which is precisely the sense in which firms scale by intensifying existing occupations rather than acquiring new ones.

Fact 3: Core products and the intensive margin of occupations. Fact 2 shows that scope buys workers, not occupations: firms expanding their product mix grow employment

mostly within a set of occupations they already hold. This fact turns to what pins that set down in the first place, and asks whether firms that share a core product[7] also share a common occupational structure, or whether each firm's occupational makeup is largely its own.

A natural way to summarize a product's occupational requirements is to pool hours across the single-product firms that report it as their core product, treating those firms as interchangeable draws from a common occupational recipe. I test this assumption using each firm's hours pooled across the full sample period, since annual occupation vectors are too sparse at the firm level to compare reliably, collecting them into a pooled occupation vector $\mathcal{O}(j)$ per firm.[8] Appendix Table A10 tests the pooling assumption directly: regressing θ_{jo} on occupation fixed effects alone leaves most of its variation unexplained ($R^2 \in [0.27, 0.34]$ across scope bins, scope likewise measured over the full sample period); interacting occupation with core product raises this to 0.60 among single-product firms and to 0.89 among firms with ten or more products, rising monotonically in scope. Core product is therefore informative of occupational structure not only among single-product firms, but increasingly so as scope widens.

If a shared core product genuinely anchors occupational structure, firms that share one should look more alike in their occupations than firms chosen at random, and the margin on which they resemble each other reveals what that anchor actually fixes. For each core-product $\times$ scope-bin cell with at least two firms, I compute the average pairwise cosine similarity of firms' pooled occupation vectors $\mathcal{O}(j)$, separately using the hour-weighted shares above and using binary occupation-presence indicators $\mathbf{1}\{\theta_{jo} > 0\}$ in place of θ_{jo}, and compare both to the average similarity of 2,000 randomly drawn firm pairs of the same scope, unconditional on core product. Figure 2 plots the result.

Firms sharing a core product are consistently more similar to each other than random pairs of the same scope, at every scope bin and on both margins, but the gap is more than twice as wide for hour-weighted vectors as for occupation presence, and it holds steady across

[7] A firm's core product is the product accounting for the largest share of its sales, computed over the full sample period.

[8] Let h_{jot} denote firm j's hours worked in occupation o in year t, $h_{jo} = \sum_t h_{jot}$ its total hours in occupation o across the sample period, and $h_j = \sum_o h_{jo}$ its total hours. Then $\theta_{jo} = h_{jo}/h_j$ and $\mathcal{O}(j) = (\theta_{j1}, \theta_{j2}, \ldots, \theta_{j|\mathcal{O}|})$.

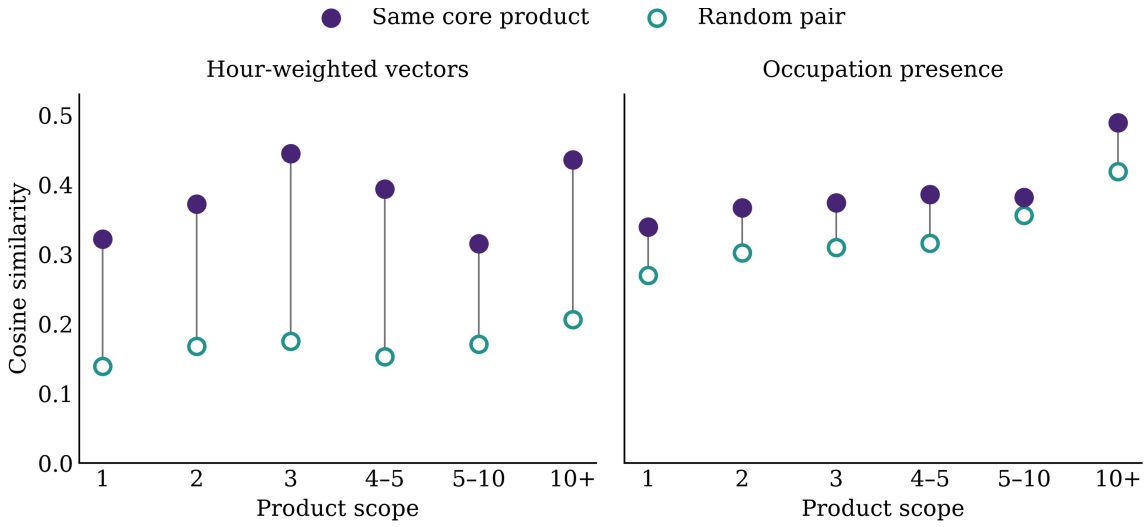


Figure 2: *Average pairwise cosine similarity of firms' occupation vectors, hours pooled across the full sample period, within core-product × scope-bin cells ("Same core product"), against a benchmark of 2,000 randomly drawn firm pairs of the same scope, unconditional on core product ("Random pair"). Left: hour-weighted occupation vectors. Right: binary occupation-presence vectors.*

scope bins rather than eroding as portfolios diversify. Taken together with the predictiveness result above, Fact 3 describes an occupational structure anchored to the firm's core product and legible mainly through the intensive margin: firms sharing a core product agree much more on how hours are allocated across occupations than on which occupations they touch at all.

3 Product-to-product similarities in occupations

Products vary in the tasks they require, and thus in the occupations needed. In this section, I aim to construct product-level task vectors, and then compute similarities between these to establish bilateral task similarities.

To do so, I will first propose a method to empirically proxy feasible tasks to produce products based on SPFs. In this context, let $\mathcal{O}(\omega)$ denote a vector of occupations (i.e. task bundles) revealed to produce any product ω, which has length $|\mathcal{O}| = 167$ equal to the unique number of three-digit occupations in the sample.

To credibly estimate the bilateral similarities of these task-vectors, I introduce an ex-

tension of the canonical cosine similarity approach that is popular in the literature (see for instance Boehm et al. (2022) and Bloom et al. (2013)): $\langle \mathcal{O}(\omega), \mathcal{O}(\omega') \rangle$. The extension is a weighting matrix of skill-sharing between bilateral occupation pairs, and it allows one to account for task-similarity between input types; i.e. that some occupations share more tasks (and skills) than others.[9] Fact 3 (Section 2.2) shows firms sharing a core product are still modestly more alike than random pairs in raw occupation presence, consistent with such near-substitutes, rather than unrelated hires, driving that residual extensive-margin overlap; the weighting scheme credits this instead of treating it as noise.

Estimated similarity structures are then illustrated across the SPF-covered product space. Distinct clusters are revealed, which display high intra-similarity and low inter-similarity. Interestingly, the clusters and individual similarities span across multiple two-digit product categories and reveal economically meaningful similarity structures.

Lastly, I report correlation statistics between the baseline estimation and alternative specifications, including alternative input metrics.

3.1 Required occupations to produce

In this section, I construct a product-level measure of the occupations feasible for production, and hence implicitly the underlying tasks, across the product space. The measure relies on observed occupational structures of firms producing within a single product category to proxy for product-level task requirements. This requires a binary classification of firms into SPFs and MPFs.

Appendix B.1 contains an elaborate data-driven approach to this. It shows that many firms produce only one product, but they cover only a small share of the (full sample) product space, introducing a trade-off between product coverage and cleanness of the associated occupational vectors. Appendix B.2 documents an alternative construction that relaxes this trade-off by testing the SPF condition at a coarser product aggregation level than the one used to assign the core product, and reports coverage and cleanness across all combinations of the two aggregation levels.

[9] At the three-digit occupation level, two types of engineers are for example included. Workers in these occupations are likely sharing more tasks than a worker in gardening, for instance.

Going forward, I treat SPFs as firms with 90 percent or more of their manufacturing sales in a single product category in every sample year. Under this definition, nearly 93 percent of total manufacturing sales across the period is covered by SPF produced products, although $\mathcal{P}$ spans less than half of four-digit product categories. Using a core-product sales threshold of 90 percent could contaminate occupational vectors if many workers are tied to different products than the SPF core product. However, within SPFs, nearly 99 percent of sales still come from the core product, suggesting little concern with the cleanness of vectors.

By extension, MPFs are defined as having less than 90 percent of their sales in the core product across all sample years. Therefore, I also exclude firms that switch status across the sample period. MPFs in the sample produce 6.4 products on average. The average sales-share concentration in the core product is 63 percent, and the mean sales share across all products is 26 percent.

A key step is then to move from the extensive margin of occupational presence to the intensive margin: Fact 3 (Section 2.2) shows that firms sharing a core product resemble each other roughly twice as much in hour-weighted occupation vectors as in occupation presence alone, so the intensive margin is where the core-product signal actually lives. The objective is therefore to measure the relative importance of each occupation in the production of a given product.

I will treat $\mathcal{O}(\omega)$ as static, and thus omit subscript t. This reflects a practical limitation of the data; SPF-covered product space $\mathcal{P}$ and the associated vectors would be noisy and unbalanced if constructed year by year. The downside of omitting the time dimension is sensitivity to time-varying changes in production technologies that affect task requirements at the product level. In all relevant regressions, I attempt to mitigate this through interacted fixed effects.

Let SPF denote the full set of SPFs. Then, for each $j \in \text{SPF}$, I first compute the total number of hours worked in each observed occupation, and subsequently group them by their core product $\bar{\omega}_j$, such that each $j \in \text{SPF}$ belongs to the set of SPFs producing $\omega = \bar{\omega}_j$. I denote the set of SPFs producing product ω by $\text{SPF}_\omega \subseteq \text{SPF}$. Pooling across firms with a common core product this way is grounded in Fact 3 (Section 2.2): interacting occupation with core product explains up to 89 percent of a firm's occupation-level hours

share, confirming that core product pins down a common occupational recipe rather than an arbitrary one.

Occupation-specific working hours at the product level are computed as the sum of hours worked in occupation o across all firms $j \in \text{SPF}_\omega$. Formally,

$$h_{\omega o} = \sum_{j \in \text{SPF}_\omega} h_{jo}, \quad \text{and} \quad h_\omega = \sum_o h_{\omega o}.$$

The relative importance of occupation o in producing product ω is then given by the share of total labor hours:

$$\theta_{\omega o} = \frac{h_{\omega o}}{h_\omega}, \quad \text{with} \quad \sum_o \theta_{\omega o} = 1.$$

This yields one product-level vector $\mathcal{O}(\omega)$ per covered product, and $|\mathcal{P}|$ vectors in total, each of length $|\mathcal{O}|$, the total number of unique occupations: 167. Each vector summarizes the occupational composition of production for a given product:

$$\mathcal{O}(\omega) = \left(\theta_{\omega 1}, \theta_{\omega 2}, \ldots, \theta_{\omega |\mathcal{O}|}\right).$$

Under the assumption that occupations proxy for the underlying tasks, these vectors can be interpreted as reduced-form representations of product-specific production technologies in task space.

Figure 3 illustrates such vectors for three hypothetical products with five occupational inputs. Products and occupations are represented along the horizontal dimensions, while the height of each column reflects the relative importance of each occupation, measured as the share of labor input hours.

In this example, occupation 1 (e.g., managers) is used in the production of all products, but is most important for product 2. Occupation 2 (e.g., specialists) is also used across all products, but most intensively in product 3. Occupation 3 (e.g., office work) is primarily used in products 1 and 3, but is also present in product 2. Occupation 4 (e.g., sales roles) is most important for product 1, but contributes to production across all products. Finally, occupation 5 (e.g., assembly work) is used only in products 1 and 2 and is absent from product 3. Numerical values for this example are provided in Appendix C.1.

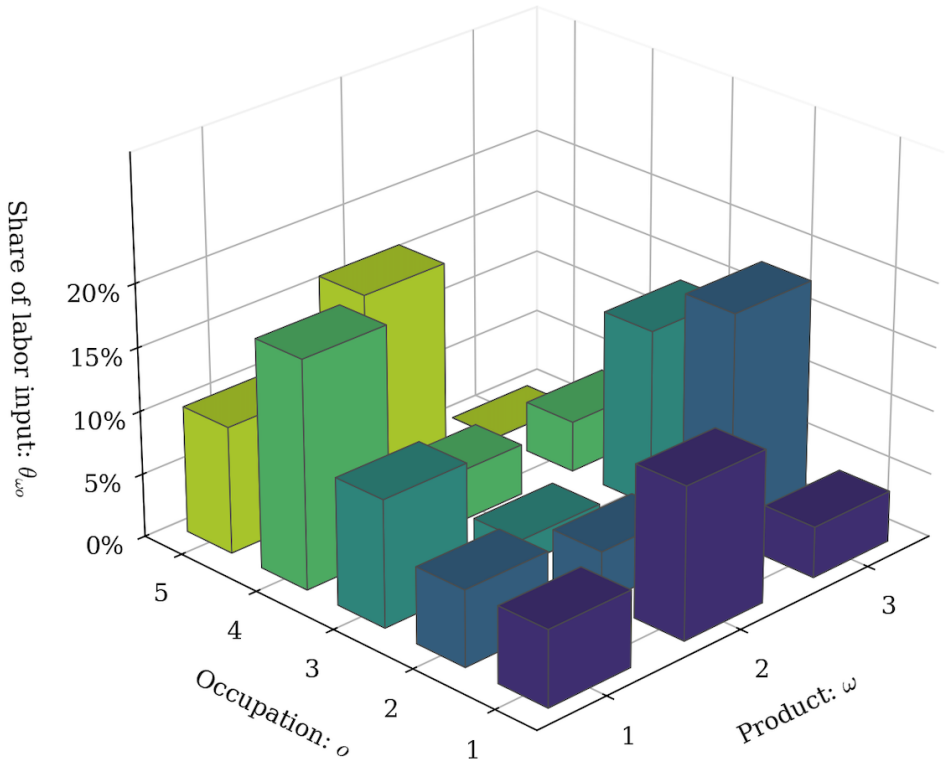


Figure 3: *Figure 3 illustrates product-level occupational vectors for three hypothetical products, each using five occupations. The height of each column reflects the share of labor input hours. All data are fictional and constructed for illustrative purposes.*

3.2 Bilateral product similarity in tasks

In this section, I will turn to the construction of bilateral product similarities. Using a standard cosine similarity measure, each occupational input share in ω is compared only to the corresponding occupation in ω'. However, not all occupations are equally distinct in terms of the tasks they perform. A simple inner product therefore does not account for heterogeneity in skill or task overlap across occupations. To address this, I introduce a weighting matrix Ψ, with element Ψ_{o,o^*} capturing the estimated skill transferability from an origin occupation o to a destination occupation o^*.

With this extension, occupations are no longer compared only one-to-one, but each occupation is compared to all other occupations, weighted by their degree of estimated skill relatedness.[10] As a result, all bilateral occupation pairs can contribute to product similar-

[10] An unweighted cosine similarity between two vectors is the sum of $|\mathcal{O}| = 167$ elements. A weighted vector is the sum of $|\mathcal{O}|^2 = 27,889$ elements. See Appendix C.1 for an example.

ity, provided that both occupational presence and skill overlap are sufficiently large. The magnitude of task similarity is therefore determined jointly by the relative importance of occupations and the degree of estimated skill transferability between them. The resulting similarity measure between products ω and ω' is defined as:

$$\Phi(\omega, \omega') = \frac{\sum_{o,o^*} \Psi_{o,o^*} \cdot \theta_{\omega o} \cdot \theta_{\omega' o^*}}{\sqrt{\sum_{o,o^*} \Psi_{o,o^*} \cdot \theta_{\omega o} \cdot \theta_{\omega o^*}} \cdot \sqrt{\sum_{o,o^*} \Psi_{o,o^*} \cdot \theta_{\omega' o} \cdot \theta_{\omega' o^*}}} \tag{2}$$

where $\theta_{\omega o}$ denotes the share of working hours assigned to occupation o among all SPFs producing product ω, following Section 3.1, such that each element corresponds to the height of the columns in Figure 3. The weighting matrix Ψ is square, with $\Psi \in \mathbb{R}^{|\mathcal{O}| \times |\mathcal{O}|}$.

Appendix C.1 provides a numerical example of both unweighted and weighted cosine similarities for the products illustrated in Figure 3. The example shows that introducing the weighting matrix not only affects the magnitude of bilateral similarities, but can also alter the ranking of similarity across product pairs.

Occupational skill-relatedness The weighting matrix, Ψ, is constructed following the approach of Neffke and Henning (2013), who estimate skill relatedness between industries using excess labor flows. I adapt this method to the occupational level, in line with Joyez and Laffineur (2022), who construct an "Occupation Space" using French data. Specifically, each element of the matrix is defined as:

$$\Psi_{o,o^*} = \frac{\mathcal{F}_{o,o^*}}{\hat{\mathcal{F}}_{o,o^*}}, \tag{3}$$

where $\mathcal{F}_{o,o^*}$ denotes the observed worker flow from an origin occupation o to a destination occupation o^*, and $\hat{\mathcal{F}}_{o,o^*}$ is the expected flow, estimated using a zero-inflated negative binomial (ZINB) regression with occupation-level characteristics as predictors.[11]

The observed flow is the number of workers who change occupational code from o to o^* during the sample period. I do not restrict the sample to workers employed in manu-

[11] Occupations with high wage growth experience larger in-flows of workers and lower out-flows, and large occupations are estimated to have a higher likelihood of any flow. Full regression results are reported in Appendix C.2.

facturing firms, as the estimation of bilateral occupation skill relatedness is independent of product-level data, and I do not distinguish between job switchers and workers who change occupational code within an ongoing employment relationship (see Groes et al., 2015, on the relative importance of the two margins). Occupational mobility rates in my sample are broadly in line with those reported for Denmark by Groes et al. (2015) and Humlum (2021a; 2021b), and for the United States by Moscarini and Thomsson (2007) and Kambourov and Manovskii (2008). Appendix C.2 documents mobility rates, the split between within- and between-firm transitions, and the direction of moves across DISCO categories.

Two features of the observed flow matrix motivate the specification in Equation 3. First, in- and outflows are asymmetric within occupation codes, which supports an asymmetric specification of Ψ_{o,o^*} (see also Etheridge et al. (2024) and). Second, bilateral flows are sparse: approximately half of all occupation pairs exhibit zero flows. The ZINB model accommodates this by jointly modeling the probability that a positive flow occurs and the size of the flow, capturing both the extensive and intensive margins of skill transferability. Appendix C.2 reports the supporting evidence, the estimation results, and the occupation pairs with the highest estimated skill transferability.

Visual depiction and product clusters The product relatedness estimates derived from Equation (2) are visualized as a symmetric similarity matrix, with products on both axes.[12] In the heatmap shown in Figure 4, brighter yellow colors indicate higher occupational similarity between product pairs, while darker blue colors indicate lower similarity.

For confidentiality reasons, not all products are included in the figure. Specifically, the matrix displays only products with more than five unique SPFs observed over the period 2010–2019 (among other restrictions), i.e., products with $|\text{SPF}_\omega| > 5$, in line with GDPR requirements. This constraint leads to the exclusion of a disproportionately large number of numerically low product categories. Although not displayed in Figure 4, a measure of Φ is obtained for all bilateral product pairs.

The heatmap reveals several clear clusters in the product space. First, the upper-left

[12]The underlying similarity matrix, i.e. the same GDPR-compliant set of product pairs shown in Figure 4, is available for download at the online appendix, along with documentation, so that readers can use it directly in their own work: `https://kristianskovisen.github.io/online-appendix/product-similarity.html`.

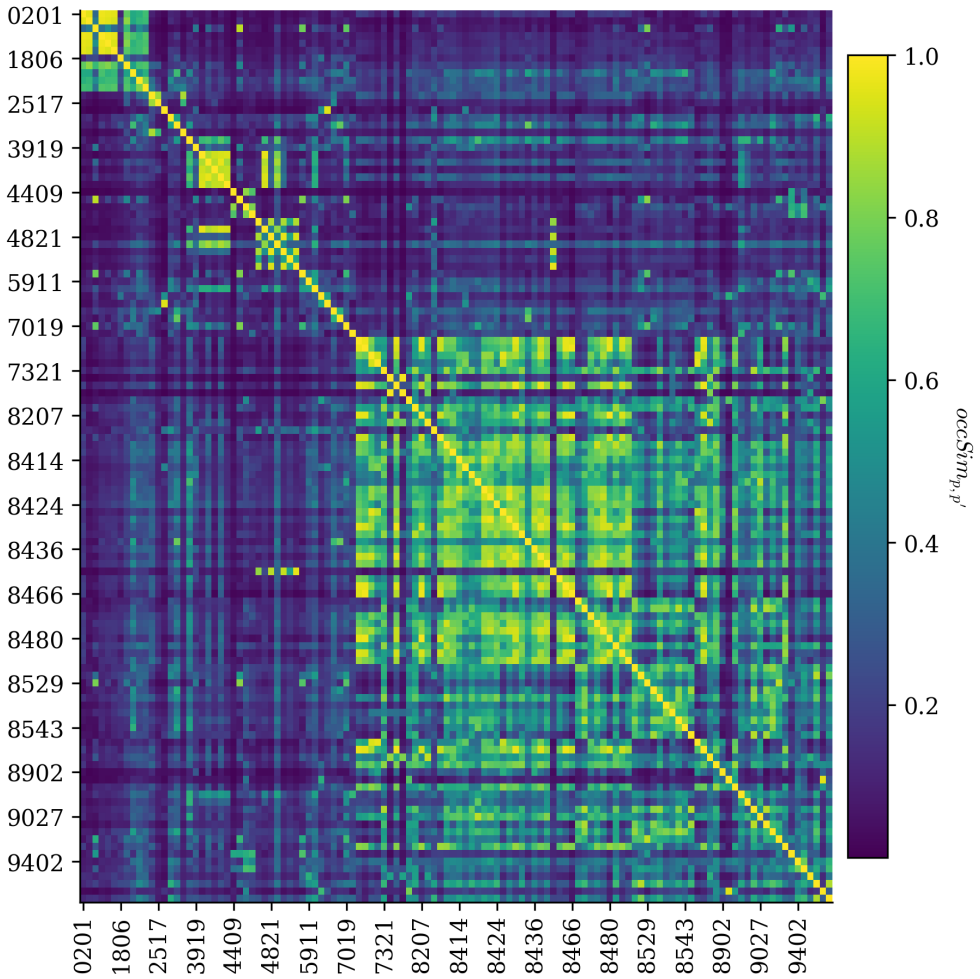


Figure 4: *Figure 4 displays product similarities in occupations, $\Phi(\omega, \omega')$, constructed from Equation 2. High similarities (closer to 1) appear in brighter yellow, while lower similarities (closer to 0) appear in darker blue. The diagonal contains values equal to 1. Only products satisfying GDPR requirements are included.*

corner contains a compact food-related cluster, with strong similarities among "0201: Meat (fresh/chilled)", "0304: Fish (fillets/fish meat)", "1601: Meat preparations", "1604: Fish preparations", and "1704: Sugar confectionery". Within this cluster, pairwise similarity values are high. For example, the similarity between "1601: Meat preparations" and "1604: Fish preparations" is approximately 0.98, while "0201: Meat (fresh/chilled)" is also highly similar to both (around 0.97 and 0.96, respectively).

Second, there is a cluster of plastic and rubber products centered around "3920: Plastics" and "4016: Rubber". This cluster also includes "3923: Plastics and Ethylene polymers", as

well as other plastic types (3925 and 3926).

The most prominent feature of the heatmap is a large machinery and equipment cluster located in the central part of the matrix, spanning roughly from "8412: Engines" to "8486: Machines and apparatus". The densest core includes products such as "8421: Centrifuges and machinery", "8422: Dishwashing machines and machinery", "8424: Fire extinguishers and spray guns", "8426: Cranes and derricks", "8428: Lifts, elevators and conveyors", and "8431: Machinery and lifts". This cluster forms the largest contiguous region of high similarity, suggesting a shared underlying capability structure.

An important feature of the figure is that high similarities do not necessarily occur between numerically adjacent product codes. For example, the occupational composition of "8707: Vehicles" differs quite substantially from that of "8708: Vehicle parts and accessories" ($\Phi(8707, 8708) = 0.489$), despite their proximity in the classification. In contrast, 8707 is more similar to products such as "8431: Machinery and lifts" or "7304: Iron or steel".[13] Other notable examples include "8207: Tools, interchangeable", which is closely related to "8480: Moulds" and "8466: Machine-tools" (both around 0.98). Similarly, "7304: Iron or steel (excluding cast iron)" is highly similar to 8480, 8466, and 8707. "7326: Iron or steel" aligns more strongly with "8431: Machinery and lifts" and "8437: Machines and machinery" than with many nearby codes.

Some isolated pairings stand out particularly clearly: "4911: Printed matter" is highly similar to "8442: Machinery, apparatus and equipment for printing matter" (approximately 0.96), "7309: Reservoirs, tanks, vats and similar containers" is similar to "8609: Containers" (approximately 0.96), and "9406: Buildings prefabricated" to "8903: Yachts and other vessels" (approximately 0.95). These off-diagonal relationships suggest that the similarity measure captures shared production and occupational requirements that cut across the HS classification, rather than simply reflecting proximity in product codes. Appendix C.3 reports the 30 most similar bilateral product pairs, which confirms the patterns observed in the heatmap.

More generally, high similarities are relatively rare, as most product pairs share limited overlap in occupational composition. Appendix C.3 also presents a kernel density plot of the

[13] $\Phi(8707, 8431) = 0.968$ and $\Phi(8707, 7304) = 0.960$.

estimated bilateral product similarities. The distribution is right-skewed and peaks around 0.2, while only a small fraction of product pairs exhibit similarity values above 0.8.

3.3 Assumptions and robustness

The main results on bilateral product task similarities rely on a number of assumptions and methodological choices by the author. In this section, I address several of the most important ones and provide brief robustness analyses which aim to inform how the task similarity measure changes with sample or measurement alterations. Table 3.3 contains correlation metrics relating the benchmark task similarity matrix illustrated in Figure 4 to each alternative construction, including Pearson r, Spearman ρ, and Kendall τ. The former measures linear correlation, and is thus indicative of whether two matrices assign the same relatedness *levels* to product pairs, whereas Spearman ρ and Kendall τ instead assess whether the two constructions agree on the *ordering* of which products are close in task space. Because ρ and τ depend only on ordering, they are invariant to any monotone rescaling of similarity. This scale invariance is particularly important for the product selection mechanism as shown in previous literature, since the product choice mechanism operates on which candidates rank closest in task space rather than on cardinal similarity; τ provides the more conservative, tie-robust reading of that ordinal agreement. Note that τ tends to be lower than r for a given strong monotone association, so the gap between the two across the high-correlation rows below is partly mechanical.

A first assessment concerns the role of the weighting matrix, corresponding to the standard cosine similarity approach used in the literature (for input-based similarities, see Boehm et al. (2022); for sector-level similarities, see Bloom et al. (2013)). Correlating their off-diagonal elements yields a Pearson r of 0.974, while the rank-based coefficients are somewhat lower ($\rho = 0.962$, $\tau = 0.860$). Omitting Ψ_{o,o^*} thus leaves the levels of product relatedness essentially intact and perturbs mainly their ordering.

A second alternative replaces the hour-weighted occupational vector with a vector of extensive-margin binary indicators of occupational presence among any SPF_ω, thereby assessing the importance of the hour-weighting in constructing $\mathcal{O}(\omega)$. One can think of this specification as a two-dimensional version of Figure 3, in which the third vertical direction is

collapsed to a 0/1 indicator function. Here both level and rank correlations with the benchmark are low ($r = 0.361$, $\tau = 0.300$), indicating that the intensive-margin hour weights in $\mathcal{O}(\omega)$ carry substantial importance. Adding Ψ_{o,o^*} to this extensive-margin construction barely moves the correlations (0.369 versus 0.361 in r).

In constructing the product-level occupational vectors, all occupations are included. Ideally, one might distinguish between workers directly involved in production and those performing administrative or managerial tasks. This gives rise to an implicit assumption: all occupations contribute to production in proportion to their employment shares. Addressing this concern empirically is challenging. One possible, but rather crude, approach is to exclude managerial occupations (DISCO group 1) when constructing the product-level vectors. The off-diagonal Pearson correlation with the benchmark is 0.952, the rank-based coefficients only somewhat lower ($\rho = 0.922$, $\tau = 0.789$). The relatedness pattern is therefore mainly driven by employment outside the managerial occupations, consistent with production- and task-specific labor, rather than administrative composition, accounting for cross-product proximity.

A further concern relates to the underlying distribution of SPF vectors within products. Even if two firms employ the same occupations, workers may not perform identical tasks across firms, which raises the question of whether occupations accurately map into task requirements. Some firms may be more likely to assign workers to tasks that differ from their formal occupation codes. I perform a brief robustness exercise in which product-level vectors are weighted by the inverse of value added per worker of the SPF, a proxy for firm-level productivity. The premise is that more productive firms have greater flexibility in assigning workers outside their core tasks and should therefore receive less weight in the construction of product-level vectors. The reweighted matrix is quite similar to the benchmark cardinally while only ordinally different to a limited degree ($r = 0.985$, $\tau = 0.916$).

Understanding occupational similarity between products is also improved when related to alternative measures. One obvious comparison is product relatedness in alternative inputs, for instance via firms' sourcing of intermediate inputs. I compute an analogous matrix using the VARK database for all SPFs in the sample, grouping SPFs by their product and constructing a vector of sourcing inputs at the four-digit product level. I then take the

Table 2: Correlation between matrix measures

Matrix	Pearson r	Spearman ρ	Kendall τ
`Unweighted cosine`	0.974	0.962	0.860
`Extensive margin occupational presence`	0.361	0.430	0.300
`Unweighted cosine, extensive margin`	0.369	0.432	0.302
`No managers`	0.952	0.922	0.789
`Inverse productivity weighting`	0.985	0.984	0.916
`Input, VARK`	0.112	0.151	0.103

Notes: Each row reports the association between two matrices based on their off-diagonal correlations. *Pearson* r measures the linear correlation between corresponding entries. *Spearman* ρ is the rank correlation, capturing any monotonic (not necessarily linear) association. *Kendall* τ is a rank correlation based on the share of concordant minus discordant pairs, and is typically more conservative in magnitude than ρ. 'Unweighted cosine' reports the similarity matrix not including Ψ_{o,o^*}: $\langle \mathcal{O}(\omega), \mathcal{O}(\omega') \rangle$; 'Extensive margin occ. presence' disregards the intensive margin presence and only considers binary presence; 'Unweighted cosine, extensive margin' disregards both hour-weighting and Ψ_{o,o^*}; 'No managers' disregards all occupations in main category 1; 'Inverse productivity weighting' weights SPFs within products by their inverse productivity measured by VA per worker; 'Input' computes an analogous matrix using sourcing information (not all products covered by input matrix). All coefficients are statistically significantly different from 0 at the 1% level ($p < 0.001$).

(unweighted) cosine similarity between these vectors to construct the 'Input' relatedness matrix. This input-based matrix correlates only weakly with the occupation-based benchmark, in both levels and ranks ($r = 0.112$, $\tau = 0.103$). An important caveat to be mentioned: VARK covers substantially larger firms, fewer SPFs, and therefore fewer products. The correlation is therefore likely to be downward biased, as the two matrices are constructed on different samples.[14] Occupational task proximity and intermediate-input proximity therefore capture largely distinct dimensions of product relatedness, as far as can be measured with VARK. Firms close in what they source are not, in general, close in the labor capabilities their production requires.

4 Scope Adjustments and Task Similarity in Theory

To formalize expectations for product change behavior on the basis of the task similarity measure, this section provides a brief extension of the MPF framework in Eckel and Neary (2010) that links firms' product scope decisions to occupational similarity across products.In

[14]Contrary to VARS, which covers firms with 10 or more employees, VARK only includes firms with more than 50 employees.

particular, I show how $\Phi(\bar{\omega}_j, \omega)$ can be interpreted as a determinant of marginal production costs, and how this feeds into firms' endogenous scope choices.

Eckel and Neary (2010) model the marginal cost of producing any product as wages (prices) multiplied by a technological component (new task requirements), which increases in the distance to a firm's core competences. Inspired by this, I let ω denote a candidate product for firm j, whose core competences are proxied by its core product $\bar{\omega}_j$.

The latent firm-level cost of adopting a product ω depends on the set of new tasks required to produce ω, which is captured by $1 - \Phi(\bar{\omega}_j, \omega)$, where $\Phi(\bar{\omega}_j, \omega) \in [0, 1]$ again denotes the similarity in tasks between the firm's core product $\bar{\omega}_j$ and the candidate product ω. The marginal cost of producing a candidate product is thus modeled as

$$c_j(\omega) = W \cdot \left[\gamma_\omega^0 + \kappa \left(1 - \Phi(\bar{\omega}_j, \omega)\right)^\eta\right]. \tag{4}$$

Here, W denotes wages, γ_ω^0 captures the baseline task requirement of ω (and may be zero), κ governs the steepness of the marginal cost function, and η determines its curvature. I maintain $\kappa > 0$ and $\eta > 0$ throughout, so that marginal cost increases in the distance to core competences; the curvature parameter η is otherwise left free and revisited in the empirical analysis. In the spirit of Eckel and Neary (2010), one can illustrate the model as displayed in Figure 4 for $\kappa > 0$ and $\eta = 1$.

The penalty term $\kappa(1 - \Phi(\bar{\omega}_j, \omega))^\eta$ is the cost-side counterpart of Fact 3 (Section 2.2), which shows that a shared core product predicts how firms weight occupations by hours far more than which occupations they employ at all. $\Phi(\bar{\omega}_j, \omega)$ is built from exactly this hour-weighted structure, so $1 - \Phi(\bar{\omega}_j, \omega)$ measures the reallocation of hours across the firm's existing occupations that producing ω would require, and $\kappa(1 - \Phi(\bar{\omega}_j, \omega))^\eta$ prices it: the more a candidate's task requirements differ in intensity from the core product's, the larger the reallocation, and the higher its marginal cost. Fact 2, in turn, is consistent with why this cost is worth paying: firms that do add products scale employment sharply within occupations they already hold, which is only profitable if the intensive reallocation required at the margin is small, i.e., if $\Phi(\bar{\omega}_j, \omega)$ is high.

Using the same aggregate demand setup as in Eckel and Neary (2010), which is outlined

in Appendix D, a candidate product ω is viable if its effective (choke) price at least covers marginal cost:

$$a' - W \cdot \left[\gamma_\omega^0 + \kappa \left(1 - \Phi(\bar{\omega}_j, \omega)\right)^\eta\right] - b' e \left(X_j + Y\right) \geq 0, \tag{5}$$

where a', b', and e are non-negative and denote the broad willingness to pay, the slope of demand (which decreases in market size), and substitutability across products, respectively. X_j is firm j's output of other products, capturing cannibalization effects, and Y denotes aggregate output of the product across all firms, capturing competition effects.

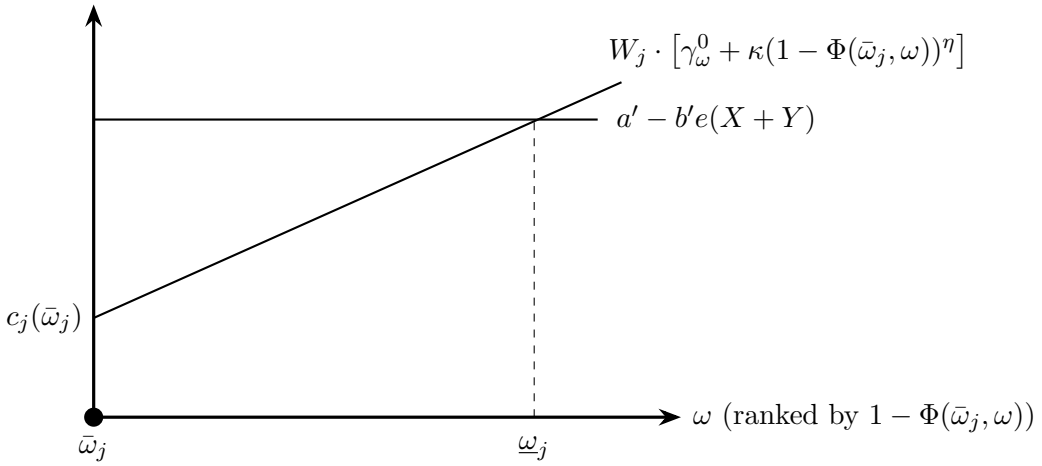


Figure 5: Authors own drawings adapted from Eckel and Neary (2010). Illustration of firm product mix decisions with flexible manufacturing: The firm's marginal costs of producing any latent product increase with distance in required tasks to the core product of the firm. A firm will produce any product where marginal costs are below marginal revenue.

For convenience, I collect the demand and cost shifters into the net surplus:

$$s_{j\omega} \equiv a' - W\gamma_\omega^0 - b' e \left(X_j + Y\right), \tag{6}$$

so that Equation (5) is equivalent to the viability condition $s_{j\omega} \geq \kappa W \left(1 - \Phi(\bar{\omega}_j, \omega)\right)^\eta$. The firm's product scope is the set of products satisfying $c_j(\omega) \leq a' - b'e(X_j + Y)$, which also defines the range of products over which firm j has cost advantages from flexible manufacturing, denoted $\left[\bar{\omega}_j, \underline{\omega}_j\right]$, where $\underline{\omega}_j$ is the marginal product.[15]

[15] Under a common baseline task requirement, $\gamma_\omega^0 \equiv \gamma^0$, the threshold below is identical across candidates and the viable set is an interval in similarity space, $\Phi \in [\Phi_j^*, 1]$, with $\underline{\omega}_j$ attaining Φ_j^*. With product-specific γ_ω^0 the cutoff varies across candidates and the viable set need not be a strict interval in similarity.

Rearranging Equation (5) yields a similarity threshold for production, all else equal, such that production requires $\Phi(\bar{\omega}_j, \omega) \geq \Phi^*_{j\omega}$.[16] This threshold is the minimum similarity required for a firm to produce a candidate product. Since dropping applies only to products already in the firm's portfolio, let $\Omega_j \subseteq \mathcal{P}$ denote firm j's current product mix within the product space $\mathcal{P}$. The adding and dropping rules are then exact mirror images, restricted to non-incumbent and incumbent products respectively:

$$\text{Add}_{j\omega} = \mathbf{1}\{\omega \notin \Omega_j\} \cdot \mathbf{1}\{\Phi(\bar{\omega}_j, \omega) \geq \Phi^*_{j\omega}\}, \tag{7}$$

$$\text{Drop}_{j\omega} = \mathbf{1}\{\omega \in \Omega_j\} \cdot \mathbf{1}\{\Phi(\bar{\omega}_j, \omega) < \Phi^*_{j\omega}\}. \tag{8}$$

The marginal product $\underline{\omega}_j$ is defined where the inequality binds. In this frictionless static benchmark a single cutoff governs both margins: a product is viable if and only if $\Phi \geq \Phi^*_{j\omega}$. The model thus abstracts from adjustment costs, and any empirical asymmetry between adding and dropping operates through differences in the joint distribution of $(\Phi, s_{j\omega})$ across the candidate and incumbent pools, rather than through separate thresholds.

To move from the deterministic rules to testable statements about probabilities, note that the demand and cost shifters entering $s_{j\omega}$ vary across firm–product pairs. I will think of the net surplus $s_{j\omega}$ as a random variable with cdf G, and equivalently the threshold $\Phi^*_{j\omega}$ is stochastic. Conditional on similarity to the core, the probability that a candidate product is added is

$$P(\text{Add}_{j\omega} = 1 \mid \Phi) = P(s_{j\omega} \geq \kappa W(1-\Phi)^\eta) = 1 - G(\kappa W(1-\Phi)^\eta)\,. \tag{9}$$

[16] Setting Equation (5) to zero and solving for Φ yields the cutoff

$$\Phi^*_{j\omega} = 1 - \left(\frac{a' - b'e(X_j + Y) - W\gamma^0_\omega}{\kappa W}\right)^{\frac{1}{\eta}} = 1 - \left(\frac{s_{j\omega}}{\kappa W}\right)^{\frac{1}{\eta}}.$$

Appendix D.5 shows that the threshold similarity is decreasing in demand and increasing in firm sales of other products, total output of the product, wages, adaptation costs, and baseline product costs. The intuition is straightforward: stronger demand lowers the effective barrier to entry, while competition, cannibalization, and cost factors raise it. The wage result relies on $s_{j\omega} > 0$, which holds for any product in the viable region.

Differentiating, the probability of adding increases in similarity to the firm's core product:

$$\frac{\partial P(\text{Add}_{j\omega} = 1 \mid \Phi)}{\partial \Phi} = g(\kappa W(1-\Phi)^{\eta})\, \kappa W \eta (1-\Phi)^{\eta-1} > 0. \tag{10}$$

In the graphical illustration in Figure 4, this implies that products closer to $\bar{\omega}_j$ in task space are more likely to be added by firm j.

The curvature of this relationship is governed by η. Because distance enters through $(1-\Phi)^{\eta}$, a steep near-core cost gradient concentrates viable products among those most similar to the core, so that adding probability rises appreciably only at the very top of the similarity distribution.[17]

Symmetrically, for incumbent products the probability of dropping is

$$P\big(\text{Drop}_{j\omega} = 1 \mid \Phi,\, \omega \in \Omega_j\big) = G(\kappa W(1-\Phi)^{\eta})\,, \tag{11}$$

which decreases in similarity:

$$\frac{\partial P\big(\text{Drop}_{j\omega} = 1 \mid \Phi,\, \omega \in \Omega_j\big)}{\partial \Phi} < 0, \tag{12}$$

implying that incumbent products closer to the marginal product in similarity are more likely to be dropped.

5 Multi-product firms and occupational similarity

In this section, I examine how occupational similarity shapes product scope decisions of MPFs, in line with the theoretical predictions outlined in Section 4. The central mechanism is that firms expand and contract their product mix along dimensions of task similarity to their core competences. I proceed in three steps. Initially, I document how existing portfolios are organized in the constructed task space presented in Section 3.2, and show that this

[17] For example, if $s_{j\omega} \sim U[0, \bar{s}]$ over the relevant range, then $P(\text{Add}_{j\omega} = 1 \mid \Phi) = 1 - (\kappa W/\bar{s})(1-\Phi)^{\eta}$ is convex in Φ; placing the mass of additions among the most similar products, if and only if $\eta < 1$. In that case the marginal cost of the first unit of distance from the core is steep, so that only products extremely close to $\bar{\omega}_j$ are viable, producing a sharp spike in adding probability for the highest similarity percentiles.

organization holds within, not only across, portfolios of a given size. Section 5.1 turns to the extensive margin of firm product scope expansion, and Section 5.2 to contraction. Both take the firm's core product as the reference point, which is the object the framework of Section 4 places at the origin of the cost function. Section 5.3 then asks whether that restriction is substantive, by replacing the core product with two aggregators, the portfolio mean and the portfolio maximum, defined over the firm's entire lagged portfolio, before turning to the static candidate pool and the alternative fixed-effects saturation as robustness checks.

Task-space organization of the product mix. The framework of Eckel and Neary (2010) organizes the firm's portfolio around a single core competence. Flexible manufacturing allows the firm to produce a range of varieties, but unit costs rise as production moves away from the core along a competence ladder, so that output and sales decline monotonically in distance from the core. The firm produces up to the marginal product $\underline{\omega}_j$, at which the cost penalty exhausts the surplus, and the viable range $[\bar{\omega}_j, \underline{\omega}_j]$ characterizes the portfolio. In the original formulation the ladder is an ordering imposed on the model: products are indexed by their distance from the core, but that distance is not measured. Section 4 adapts the framework by letting the cost penalty be governed by task distance, $1 - \Phi(\bar{\omega}_j, \omega)$, which is observed. This delivers a testable implication that the original ladder does not: if the adapted cost structure describes the data, then the position of a product in the firm's sales ranking should map systematically into its measured task similarity to the core product.

Figure 6 plots the average task similarity of products to the firm's core product as a function of product rank, ordered by sales shares. The figure reveals a clear negative relationship. Products with lower sales shares are systematically less similar to the firm's core product. This is the empirical counterpart of the competence ladder: the descent in similarity toward the marginal product $\underline{\omega}_j$ traces the ordering that Eckel and Neary (2010) assume, and recovers it from measured task content rather than imposing it. Firms concentrate production in products closest to their existing capabilities, while more distant products play a marginal role. Although product-level productivity is not directly observed, this evidence indicates that production scale is increasing in task similarity to core competences. The relationship is descriptive: portfolio composition and sales shares are jointly determined, and the figure

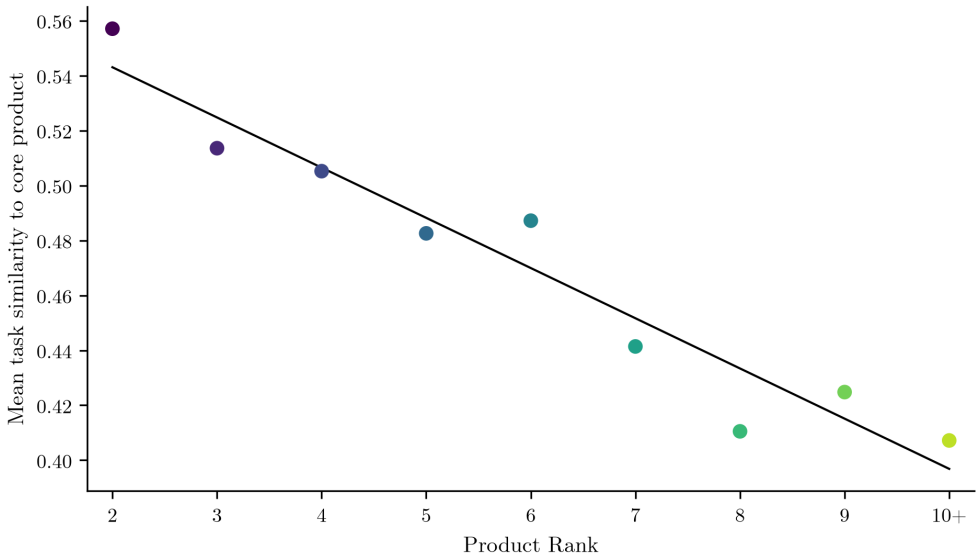


Figure 6: *This figure displays the mean task similarity (across firms and years) of co-produced products by rank. Since MPFs vary in product scope, the number of firms contributing to the mean decreases with rank. The black line shows a linear least squares fit.*

is intended to establish that the measured similarity ordering is economically meaningful before it is used to study scope adjustments.

Figure 6 pools firms of every portfolio size, so the negative relationship it documents could in principle be an artifact of composition: if larger portfolios both rank more products and happen to be less coherent, pooling could manufacture a downward slope that holds for no individual firm. Appendix Figure E1 rules this out by repeating the same exercise separately within each portfolio scope, from two to ten-or-more distinct products: similarity to the core product declines in rank within every scope group, not only in the pooled sample, confirming that the competence-ladder pattern is a within-firm regularity and not a byproduct of averaging over firms of different sizes.

5.1 Product adding

I now turn to product scope adjustments, and start by showing that product adding decisions are significantly associated with task similarity to the firm's existing competences. All adding regressions are estimated at the firm–year–candidate-product level. Firm–candidate proximity is measured by task similarity to the core product, $\Phi(\bar{\omega}_j, \omega)$, which is the object

entering the cost function in Equation (4) directly: the framework of Section 4 places the entire cost penalty on distance from $\bar{\omega}_j$, and the adding rule in Equation (7) is stated in exactly these terms. The regressions below are therefore the direct empirical counterpart of that condition. Section 5.3 asks whether the restriction to the core product is substantive, by allowing capabilities embodied in smaller incumbent products to enter as well.

Candidate pool. A product is at risk of being added only if the firm does not already produce it, so that adding contracts the pool and dropping expands it,

$$\mathcal{C}_{jt}^{D} = \mathcal{P} \setminus \Omega_{j,t-1}, \tag{13}$$

where $\mathcal{P}$ denotes the full product space covered by SPFs and $\Omega_{j,t-1}$ the firm's lagged portfolio.[18] Thus, the composition of the candidate space is itself an outcome of past scope decisions. Section 5.3 also reports results under an alternative *static* pool that instead holds the risk set fixed at the full product space in every firm–year; $\mathcal{C}_{jt}^{S} = \mathcal{P}$, which keeps the estimation sample balanced across firms and years, at the cost of retaining observations that cannot mechanically be added, and bounds the role of this choice.

Linear specification. Inspired by Boehm et al. (2022), I estimate

$$P(\text{Add}_{j\omega,t}) = \beta\, \Phi(\bar{\omega}_j, \omega) + \lambda_{jt} + \lambda_{\omega t} + \varepsilon_{j\omega,t}, \qquad \omega \in \mathcal{C}_{jt}^{D}. \tag{14}$$

The specification includes firm-by-year and candidate-product-by-year fixed effects. The former absorbs firm-specific propensities to expand together with any firm-level shock to scale, employment, or demand in a given year; the latter absorbs the general attractiveness of a candidate product in that year, including product-level demand and price movements common to all firms. The coefficient β is therefore estimated from variation across firms within a candidate-product–year cell: among firms facing the same candidate in the same year, those whose core product is closer to it in task space are more likely to add it. Core-

[18] If any candidate product ω' is added by firm j in period $t-1$, it is not included in the pool of candidate products for firm j in t; if ω' is subsequently dropped in t by j, it re-enters the list of candidate products for j in $t+1$.

Table 3: Product adding and task similarity to the core product

	(1)	(2)	(3)	(4)
	Dependent variable: $\text{Add}_{j\omega,t}$			
$\Phi(\bar{\omega}_j, \omega)$	0.0023***	0.0028***	0.0033***	0.0020***
	(0.0003)	(0.0004)	(0.0004)	(0.0005)
Fixed effects				
Firm $j \times$ Year t	No	Yes	Yes	Yes
Candidate product $\omega \times$ Year t	No	No	Yes	No
Core sector $\times$ candidate product $\omega \times$ Year t	No	No	No	Yes
Observations	1,804,240	1,804,240	1,804,240	1,804,240
R^2	0.0004	0.0085	0.0114	0.1661
Within R^2	–	0.0005	0.0005	0.0001

Notes: Sample is the dynamic candidate pool $\mathcal{C}^D_{jt} = \mathcal{P} \setminus \Omega_{j,t-1}$ of Equation (13), estimated at the firm–year–candidate-product level. $\Phi(\bar{\omega}_j, \omega)$ is task similarity between the candidate product and the firm's core product. Column (1) reports the unconditional correlation; column (2) adds firm-by-year fixed effects; column (3) is the baseline specification of Equation (14), adding candidate-product-by-year fixed effects; column (4) replaces the candidate-product-by-year fixed effects with candidate-product-by-core-sector-by-year fixed effects, saturating for the interaction between the candidate and the firm's core sector. Standard errors, two-way clustered by firm and candidate product, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

product fixed effects are redundant in this specification, as they are nested in λ_{jt}; for the same reason, firm-by-candidate fixed effects cannot be included alongside $\Phi(\bar{\omega}_j, \omega)$, which is time-invariant within firm–candidate pairs. Section 5.3 returns to this constraint, which does not bind for time-varying measures of proximity. Here, I relate to the FE specification in Boehm et al. (2022), which includes candidate-core-time fixed effects.[19] Standard errors are two-way clustered by firm and candidate product throughout, since all firms draw candidates from the same product space.

Table 3 reports the results. In the baseline specification of column (3), a one-unit increase in task similarity to the core product is associated with a 0.33 percentage point higher probability of adding a candidate product, relative to a mean adding probability under the dynamic pool of roughly 0.06 percent; the coefficient is stable in sign and significance as fixed effects are progressively saturated in columns (1)–(4). This finding is consistent with the theoretical mechanism: firms expand along dimensions where marginal production

[19] One technical difference from the fixed effects specification in Boehm et al. (2022) is the aggregation level of the core-industry. They specify the candidate and core products at the same aggregation level, which is not possible here, as this would absorb the variation from the core-to-candidate similarity completely. Therefore, I aggregate this to the core-sector, which is measured at the two-digit level.

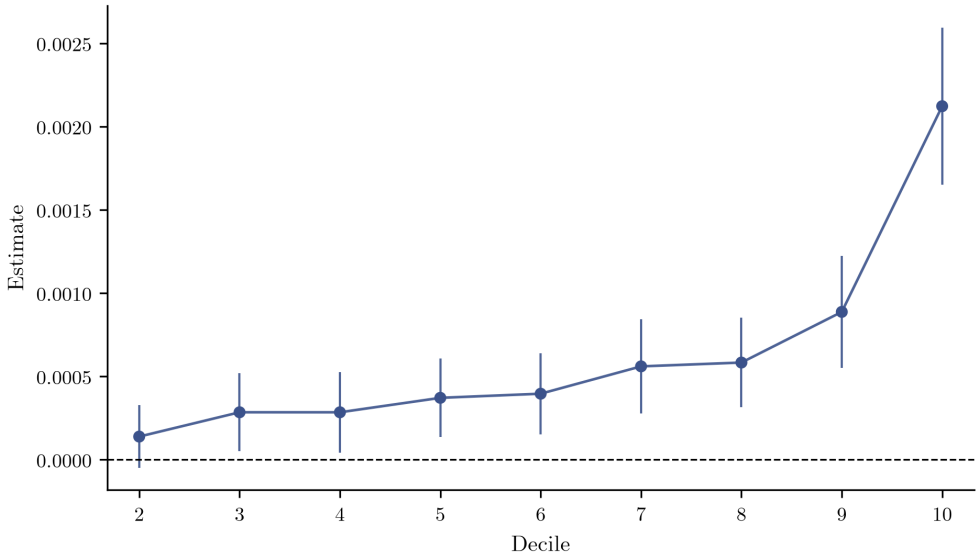


Figure 7: *This figure displays the probability of product addition by decile of task similarity to the core product, estimated on the dynamic candidate pool. The lowest decile is the omitted reference category. Vertical lines are 95% confidence intervals from standard errors two-way clustered by firm and candidate product.*

costs are plausibly lower given overlap in task requirements. In column (4), fixed effects furthermore absorb all variation in the candidate-product–year cell. The results are still positively significant, implying that even among firms who predominantly produce within the same broad two-digit chapter, both facing the same candidate ω in the same year, those whose specific four-digit core product happens to be closer to ω in task space are more likely to add it. Appendix Table E3 repeats the same specifications under the static candidate pool, with materially unchanged results.

Decile specification. The linear specification imposes a constant slope on the adding probability, whereas the adapted framework of Section 4 predicts that the adding probability is a nonlinear function of similarity, with curvature governed by the cost-curvature parameter η. As a first look at this shape, I group candidate products into deciles of similarity to the core product and re-estimate Equation (14) with decile dummies in place of $\Phi(\bar{\omega}_j, \omega)$, keeping the fixed effects of column (3).

The decile profile in Figure 7 is monotonically increasing and convex: estimates are indistinguishable from zero in the bottom two deciles, rise gradually through the middle of

the distribution, and more than double between the ninth and tenth deciles alone. This is the empirical counterpart of the cost-curvature parameter η: the adding probability takes the form $P(\text{Add}_{j\omega} = 1 \mid \Phi) = 1 - G(\kappa W(1-\Phi)^{\eta})$ in Equation (9), and a steep near-core cost gradient, formally $\eta < 1$, renders this function convex in Φ, concentrating viable additions among the candidates most similar to the core. The sharp increase concentrated in the top decile is consistent with such curvature operating jointly with the stochastic threshold $\Phi^{*}_{j\omega}$ generated by demand and cost heterogeneity across firm–product pairs.

Appendix E.2 reports the same specification at percentile resolution (Equation (33) and Figure E2), which sharpens this picture: the adding probability rises significantly across nearly the full similarity distribution but accelerates sharply at the very top, reaching roughly 0.6 percentage points in the top percentile alone. That isnearly twice the 0.33 percentage point average linear effect of Table 3 and close to three times the top-decile average above. Appendix Figure E4 repeats the decile exercise under the static candidate pool, with a materially identical profile, and Appendix Table E1 reports the underlying decile point estimates.

5.2 Product dropping

Turning to product dropping decisions, I estimate an analogous specification, with the outcome measured in $t+1$ and all regressors in t:

$$P(\text{Drop}_{j\omega,t+1}) = \beta\,\Phi(\bar{\omega}_j, \omega) + \lambda_{jt} + \lambda_{\bar{\omega}_j} + \lambda_{\omega t} + \varepsilon_{j\omega,t+1}, \qquad \omega \in \Omega_{jt}. \tag{15}$$

The estimation sample is the firm's incumbent portfolio, which changes dynamically, and the outcome is coded one period ahead: for any firm j–year t–product ω combination, $\text{Drop}_{j\omega,t+1}$ indicates that ω is not observed in the following period. The core-product measure is the interpretable object on the dropping side, for reasons set out in Section 5.3.

The results, reported in Table 4, indicate a strong negative relationship between task similarity and product dropping. In the baseline specification of column (3), a one-unit increase in similarity is associated with a 6.1 percentage point lower probability of dropping a product, attenuating to 3.9 points but remaining highly significant once the most saturated fixed effects are added in column (4). This effect is economically large. For reference, Figure

Table 4: Product dropping and task similarity to the core product

	(1)	(2)	(3)	(4)
	Dependent variable: $\text{Drop}_{j\omega,t+1}$			
$\Phi(\bar{\omega}_j, \omega)$	-0.0878***	-0.0674***	-0.0607***	-0.0392***
	(0.0088)	(0.0072)	(0.0070)	(0.0102)
Fixed effects				
Firm $j \times$ Year t	No	Yes	Yes	Yes
Incumbent product $\omega \times$ Year t	No	No	Yes	No
Core sector $\times$ incumbent product $\omega \times$ Year t	No	No	No	Yes
Observations	15,056	15,056	15,056	15,056
R^2	0.0173	0.2493	0.4436	0.7848
Within R^2	–	0.0088	0.0063	0.0020

Notes: Sample is the firm's incumbent portfolio Ω_{jt}, with the outcome coded one period ahead as in Equation (15). $\Phi(\bar{\omega}_j, \omega)$ is task similarity between the incumbent product and the firm's core product. Column (1) reports the unconditional correlation; column (2) adds firm-by-year fixed effects; column (3) adds incumbent-product-by-year fixed effects; column (4) replaces the incumbent-product-by-year fixed effects with incumbent-product-by-core-sector-by-year fixed effects. Standard errors, two-way clustered by firm and product, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

6 shows that the difference in similarity between highly ranked and low-ranked products is approximately 0.2, implying a substantial change in dropping probabilities across the product distribution. Appendix Table E9 shows that this result is not driven mechanically by the core product itself, which is dropped with near-zero frequency: excluding it, the coefficient retains its sign throughout and remains significant up to column (2).

For incumbent products, similarity levels are compressed and relatively high, so rank rather than the similarity level carries much of the economically relevant within-portfolio variation. Figure 8 reports a rank-based specification in which incumbents are grouped by their sales rank within the portfolio, using the same fixed effects as column (3) of Table 4.

The profile is monotonically increasing: the second-ranked product is statistically indistinguishable from the reference category, rising to a significant 3.8 percentage points at the third rank and to nearly 27 percentage points for the tenth-ranked product and beyond. Since lower-ranked products are also less similar to the firm's core product, this reinforces the conclusion that firms exit products that are both less important and more distant in task space. Appendix Table E8 reports the underlying point estimates.

Taken together, the adding and dropping results show that firms adjust their product

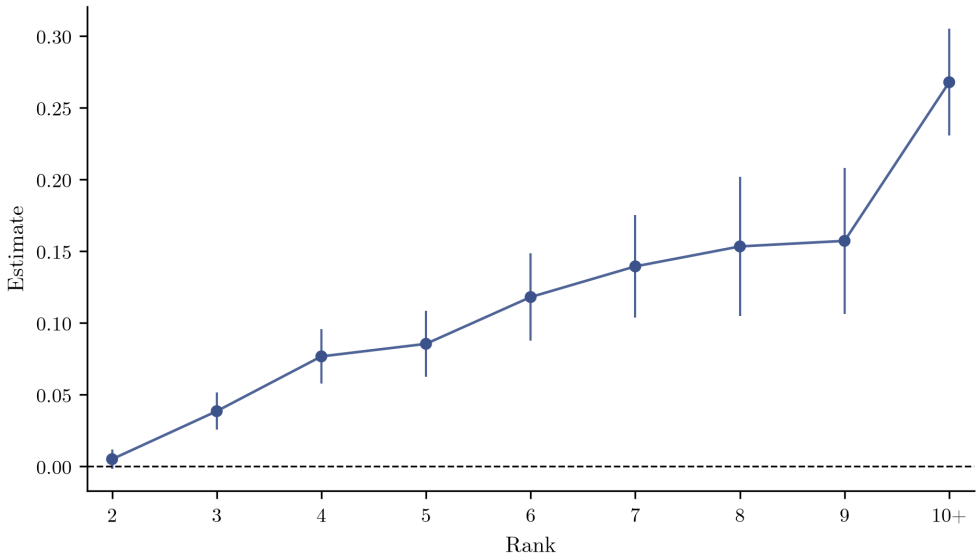


Figure 8: *This figure displays the probability of product dropping by sales rank within the portfolio. The top-ranked (typically core) product is the omitted reference category. Vertical lines are 95% confidence intervals from standard errors two-way clustered by firm and product.*

portfolios asymmetrically: they expand into products that are similar to their core competences and exit products that are more distant in task space. The findings are consistent with a model in which firms face increasing costs of adapting to distant products and therefore optimally concentrate production around their core competences.

5.3 Portfolio-based similarity

The specifications above summarize firm–candidate proximity by task similarity to the core product alone, which implicitly assumes that the capabilities relevant for expansion are exhausted by $\bar{\omega}_j$. This is the assumption the framework of Section 4 makes, but this restriction could be substantive: a candidate may be distant from the core product yet close in task space to a smaller incumbent product from which occupations can be redeployed. This section relaxes that restriction along two dimensions, the aggregator used to summarize proximity and the composition of the candidate pool, and shows that the adding results are unaffected.

Alternative aggregators. I consider two aggregators over the entire lagged portfolio,

$$\Phi^{\text{port}}_{j\omega,t-1} = \sum_{\omega' \in \Omega_{j,t-1}} \tilde{\sigma}_{j\omega',t-1}\, \Phi(\omega', \omega), \tag{16}$$

$$\Phi^{\max}_{j\omega,t-1} = \max_{\omega' \in \Omega_{j,t-1}} \Phi(\omega', \omega), \tag{17}$$

where $\tilde{\sigma}_{j\omega',t-1} = \sigma_{j\omega',t-1} \big/ \sum_{\omega'' \in \Omega_{j,t-1}} \sigma_{j\omega'',t-1}$ are renormalized lagged sales shares. The portfolio measure weights every incumbent product by its economic importance; the maximum measure asks only whether *some* incumbent product is close, and is therefore the natural test of whether redeployment operates through the single nearest capability rather than the portfolio as a whole. The core-product measure used in Sections 5.1 and 5.2,

$$\Phi^{\text{core}}_{j\omega,t-1} = \Phi(\bar{\omega}_j, \omega), \tag{18}$$

is the degenerate case of Equation (16) in which the portfolio collapses to the core product. Under the static pool, incumbent products enter the sample as candidates, so Equations (16) and (17) are computed leave-one-out, excluding ω itself from $\Omega_{j,t-1}$; under the dynamic pool of Equation (13) this correction is redundant by construction.

Linear specification. Letting $m \in \{\text{core}, \text{port}, \max\}$ index the similarity aggregator and $c \in \{D, S\}$ the candidate pool, the linear specification of Equation (14) generalizes to

$$P(\text{Add}_{j\omega,t}) = \beta^{m,c}\, \Phi^{m}_{j\omega,t-1} + \lambda_{jt} + \lambda_{\omega t} + \varepsilon_{j\omega,t}, \qquad \omega \in \mathcal{C}^{c}_{jt}. \tag{19}$$

The case $(m, c) = (\text{core}, D)$ recovers Equation (14), already reported in Table 3. I re-estimate Equation (19) for $(m, c) = (\text{port}, D)$ and $(\max, D)$ on the dynamic candidate pool, keeping the same firm-by-year and candidate-product-by-year fixed effects.

Tables 5 and 6 show that both alternative aggregators yield the same pattern as the core-product measure in Table 3: a positive, highly significant relationship between similarity and adding that survives progressively saturated fixed effects. In the baseline specification of column (3), the portfolio-mean measure is associated with a 0.39 percentage point higher

Table 5: Product adding and portfolio task similarity

	(1)	(2)	(3)	(4)
	Dependent variable: $\text{Add}_{j\omega,t}$			
$\Phi^{\text{port}}_{j\omega,t-1}$	0.0025***	0.0030***	0.0039***	0.0023***
	(0.0004)	(0.0004)	(0.0005)	(0.0007)
Fixed effects				
Firm $j \times$ Year t	No	Yes	Yes	Yes
Candidate product $\omega \times$ Year t	No	No	Yes	No
Core sector $\times$ candidate product $\omega \times$ Year t	No	No	No	Yes
Observations	1,804,240	1,804,240	1,804,240	1,804,240
R^2	0.0004	0.0085	0.0115	0.1661
Within R^2	–	0.0005	0.0006	0.0001

Notes: Sample is the dynamic candidate pool $\mathcal{C}^D_{jt}$ of Equation (13). $\Phi^{\text{port}}_{j\omega,t-1}$ is the sales-share-weighted mean task similarity between the candidate product and the firm's lagged portfolio (Equation (16)), estimated on the dynamic pool as in Equation (19) with $(m, c) = (\text{port}, D)$. Column (1) reports the unconditional correlation; column (2) adds firm-by-year fixed effects; column (3) adds candidate-product-by-year fixed effects; column (4) replaces the candidate-product-by-year fixed effects with candidate-product-by-core-sector-by-year fixed effects. Standard errors, two-way clustered by firm and candidate product, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

adding probability per unit of similarity and the maximum measure with 0.31 points, both comparable in magnitude to the 0.33-point core-product estimate.

Decile specification. The decile specification underlying Figure 7 generalizes analogously,

$$P(\text{Add}_{j\omega,t}) = \sum_{d=2}^{10} \beta_d^{m,c} \, \mathbf{1}\{\Phi^m_{j\omega,t-1} \in \mathcal{Q}_d\} + \lambda_{jt} + \lambda_{\omega t} + \varepsilon_{j\omega,t}, \qquad \omega \in \mathcal{C}^c_{jt}, \tag{20}$$

where $\mathcal{Q}_d$ denotes the d-th decile bin and the lowest bin is the omitted reference category. The case $(m, c) = (\text{core}, D)$ is Equation (20) already reported in Figure 7; Figure 9 below reports (port, D) and $(\max, D)$ alongside it.

The three profiles in Figure 9 overlap closely across the full support, with the largest divergence at the very top of the distribution, where the core-product measure is noisiest because fewer candidates achieve near-perfect similarity to a single reference product. The portfolio and maximum measures reproduce both the level and the shape of the core-product result, indicating that the restriction to $\bar{\omega}_j$ in Section 4 does not discard a distinct empirical source of relatedness: whichever object is used to summarize a firm's existing capabilities,

Table 6: Product adding and maximum task similarity to the portfolio

	(1)	(2)	(3)	(4)
	Dependent variable: $\text{Add}_{j\omega,t}$			
$\Phi^{\max}_{j\omega,t-1}$	0.0032***	0.0024***	0.0031***	0.0016***
	(0.0004)	(0.0003)	(0.0004)	(0.0005)
Fixed effects				
Firm $j \times$ Year t	No	Yes	Yes	Yes
Candidate product $\omega \times$ Year t	No	No	Yes	No
Core sector $\times$ candidate product $\omega \times$ Year t	No	No	No	Yes
Observations	1,804,240	1,804,240	1,804,240	1,804,240
R^2	0.0010	0.0085	0.0115	0.1661
Within R^2	–	0.0005	0.0006	0.0001

Notes: Sample is the dynamic candidate pool $\mathcal{C}^D_{jt}$ of Equation (13). $\Phi^{\max}_{j\omega,t-1}$ is the maximum task similarity between the candidate product and any product in the firm's lagged portfolio (Equation (17)), estimated on the dynamic pool as in Equation (19) with $(m, c) = (\max, D)$. Column (1) reports the unconditional correlation; column (2) adds firm-by-year fixed effects; column (3) adds candidate-product-by-year fixed effects; column (4) replaces the candidate-product-by-year fixed effects with candidate-product-by-core-sector-by-year fixed effects. Standard errors, two-way clustered by firm and candidate product, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

expansion tracks it in essentially the same way. This is consistent with Fact 3 (Section 2.2), where core-product predictiveness of occupational structure rises, rather than falls, with scope: if the core product remains informative even for wide-scope firms, it is unsurprising that summarizing capabilities by the whole portfolio changes little. The percentile specification of Equation (33) generalizes in the same way, replacing $\mathcal{Q}_d$ with the finer bin $\mathcal{Q}_p$; Appendix E.2 states this case as Equation (34), and Appendix Figure E3 reports the resulting percentile-level comparison, confirming the same pattern. Appendix Tables E3–E5 and Figures E4–E5 report the remaining three combinations, $(m, c) \in \{\text{core}, \text{port}, \max\} \times \{S\}$, under the static pool, all qualitatively identical to their dynamic-pool counterparts.

The dropping side. The alternative aggregators of Equations (16) and (17) are uninformative for dropping. For an incumbent product, the leave-one-out maximum is nearly mechanical, since portfolios are themselves assembled around task proximity, while the leave-one-out portfolio mean is computed over only a handful of products rather than the full space. The core-product measure therefore remains the interpretable object on the dropping side, and Equation (15) is not re-estimated with the alternatives.

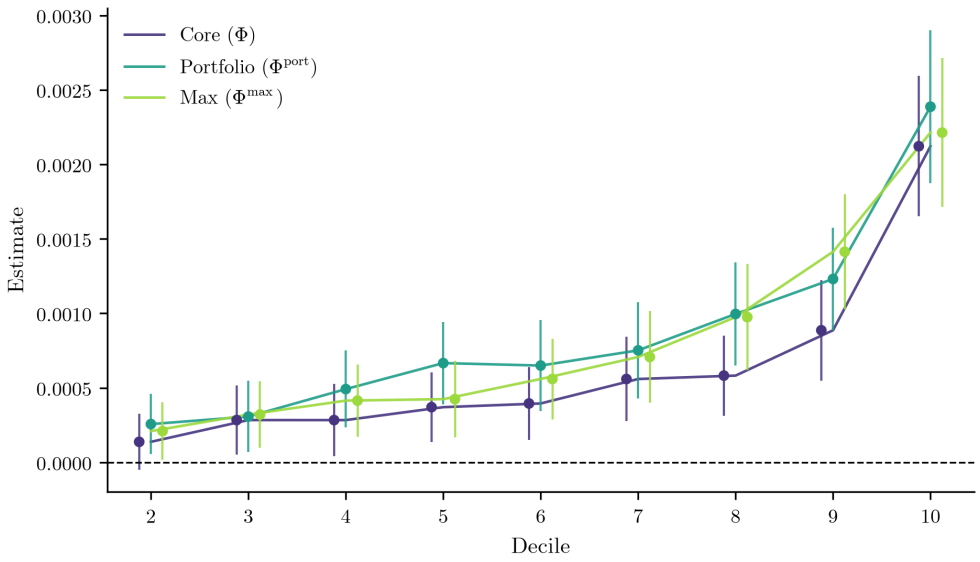


Figure 9: *This figure displays the probability of product addition by decile of similarity to the core, maximum, and portfolio-mean measures, estimated on the dynamic candidate pool. Vertical lines are 95% confidence intervals from standard errors two-way clustered by firm and candidate product.*

6 Conclusion

This paper studies how MPFs organize and adjust their product scope through the lens of occupational similarity. The central idea is that products differ in the tasks required for production, and that firms face lower costs when expanding into products that draw on capabilities already embedded in their product portfolio. By linking detailed employer–employee data to firm-level product information, the paper develops a task-based measure of product relatedness based on firms producing within only one product category and uses it to analyze product scope decisions for firms producing across multiple categories.

Taken together, the results highlight a clear mechanism of firm growth: MPFs expand in products that overlap task capabilities embodied in their current product portfolio. They retract in scope by dropping products that are less similar to their core competences. This analysis provides a novel way of understanding resource-based product diversification and supports the view that the reusability of existing capabilities shapes firm scope. More broadly, the approach developed in this paper offers a framework for studying production structure using worker-level data, and can be applied to a range of questions related to firm

dynamics, technological change, and the organization of production.

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Appendix A Data Construction and Sample Summary

A.1 Sample construction and variable definitions

This appendix collects the construction choices underlying the sample described in Section 2.1.

Firm universe and sample restrictions. Firm-level data cover all firms registered in the Danish *Dansk Branchekode 2007* sectors "B Råstofvinding" (Mining and quarrying) and "C Industri" (Manufacturing) in the VARS database. VARS contains quarterly sales for all such firms at up to ten digits of the Combined Nomenclature (CN). Firms enter only if they employ more than ten workers, measured in the third quarter of the year preceding the reference year. I exclude all firms registered within Mining and quarrying, then aggregate product-level sales to the yearly level for each firm. Firm-year information on revenue, value added, and employment is merged from the FIRM database. Because a small number of firms report employment or revenue far above the sample mean, I remove observations reporting revenue or employee counts more than three standard deviations above the yearly mean in any sample year. Firms with product counts more than three standard deviations above the yearly mean are removed on the same criterion. This analysis is not concerned with the decision and effects of product scope adjustments at the far tails of the scale and scope distributions of firms.

Product codes, concordance, and deflation. CN codes correspond to Harmonized System (HS) codes at the six-digit level. The main sample uses four-digit HS codes; the analysis can be applied at a finer level if deemed useful. As an example of the additional detail available at six digits, "0203: Meat of swine" separates into "fresh or chilled" (020319) and "frozen" (020329), a distinction I regard as too specific for the present analysis. At four digits, by contrast, "0203: Meat of swine" is separated from "0204: Meat of sheep and goats," which share the two-digit category "02: Meat." Updated HS vintages were introduced in 2012 and 2017; I handle the resulting breaks using concordance tables and convert all HS codes to their HS 2007 equivalents. All monetary values from VARS and FIRM are converted to real

values using consumer price index normalization to 2015, following Statistics Denmark.[20]

Worker sample. Information on workers in manufacturing firms is drawn primarily from the IDA databases, including IDAP, IDAN, and INDH, and linked to VARS through a unique firm identifier and year. Each worker can therefore be assigned to one or more employers, and changes in employment relationships are observed. The main sample retains workers aged 25 to 60, and only worker-year observations carrying the highest DISCO quality flag.

Occupational classification. At the worker-year level, occupations are observed as six-digit DISCO codes. DISCO is the Danish equivalent of the ISCO classification and is intended to reflect the tasks workers perform. Codes are grouped into nine main civilian categories denoted by their first digit.[21] These cover group 1, "Management"; group 2, "Work Requiring High-Level Skills, Including Research" (for instance medical doctors, engineers, and lawyers); group 3, "Work Requiring Medium-Level Skills" (for instance functions within sales, business services, and administration); group 4, "Regular Office and Service Work" (for instance registration work); group 5, "Service and Sales" (for instance customer service roles); group 6, "Workers in Farming, Forestry, and Fishing" (including gardeners); group 7, "Artisanal Work" (including carpenters, construction workers, and workers in manufacturing, excluding machine operators); group 8, "Operator and Inspector Work," including transportation and machine operators; and group 9, "Other Manual Work," including cleaning, packing, and work from groups 6, 7, and 8 with no requirement for trained skills.

At two digits the classification spans XXX unique values in the sample, including "21: Work in Science and Engineering," "31: Technical Work in Science, Engineering, Shipping and Aviation," and "81: Operator Work at Stationary Plants and Machines." At three digits, code "21" expands into six sub-codes, including "211: Physical/Earth Scientists," "212: Math/Actuarial/Statistical Professionals," "214: Engineering Professionals," and "215: Electrotechnology Engineers"; other examples are "311: Physical/Engineering Technicians" and "812: Metal Processing Operators." The sample contains XXX unique three-digit codes. At four digits the variable covers XXX unique occupations, expanding

[20] `https://www.statistikbanken.dk/pris113`

[21] A tenth main category, 0, covers military occupations.

the three-digit codes further — "413: Text and Data Processing" divides into "4131: Text processing" and "4132: Data registration work" — a level of detail I regard as unnecessary here. The main sample therefore uses three-digit DISCO occupations. A full list of DISCO codes across aggregation levels is available from Statistics Denmark.[22] Restricting the sample period to 2010 onward avoids breaks in the DISCO variable; all codes are kept in the DISCO-08 vintage. An updated version was introduced in 2012, but the revisions affect only codes more detailed than four digits.

Table A1 quantifies the non-civilian codes excluded from the three-digit civilian classification, i.e. codes registering a 0 in the tenth main category described above. Under the extensive-margin definition (Panel A), 2.52 percent of workers register a non-civilian code in at least one sample year, accounting for a similar share of total wages (3.14 percent) and hours (3.59 percent). Restricting to workers who are non-civilian throughout their entire observed spell (Panel B) leaves just 2,043 workers, or 0.55 percent of the sample; this group's wage and hours shares are smaller still, at 0.17 and 0.20 percent respectively, confirming that non-civilian codes carry negligible economic weight in the sample despite their retention (cf. Section 2).

Table A1: Non-civilian occupation shares

	Workers	Share of workers	Share of wages	Share of hours
Panel A: Non-civilian in at least one sample year				
Civilian	362,898	0.9748	0.9686	0.9641
Non-civilian	9,382	0.0252	0.0314	0.0359
Panel B: Non-civilian in every sample year				
Civilian	370,237	0.9945	0.9983	0.9980
Non-civilian	2,043	0.0055	0.0017	0.0020

Notes: A worker-year is flagged non-civilian if the third digit of its DISCO occupation code is 0 (see footnote in Section A.1 on the military main category). Panel A classifies a worker as non-civilian if this holds in at least one observed sample year (the extensive margin); Panel B restricts the non-civilian group to workers for whom this holds in every observed year, i.e. workers who are never observed in a civilian occupation. "Share of workers" reports the share of the worker sample in each row, and "Share of wages" and "Share of hours" the corresponding shares of total wage sums and total hours worked.

[22] `https://www.dst.dk/da/Statistik/dokumentation/nomenklaturer/disco-loen`

A.2 Firm and product summary statistics

This appendix reports summary statistics on firms and products (see Section 2.1). Table A2 gives an overview: 3,560 unique firms are included, with an average of 84.6 employees, and the average firm produces in 1.6 two-digit, 2.3 four-digit, and 2.9 six-digit product categories.

Table A2: Firm-level summary statistics

Firms	Mean revenue	Mean value added	Mean employees	Mean years worked	Mean products (HS2)	Mean products (HS4)	Mean products (HS6)
3,560	158.5M	51.4M	84.6	66.5	1.61	2.28	2.91

Notes: The table reports average firm characteristics across the sample, including revenue, value added, employment, years worked, and product scope across HS aggregation levels.

The mean masks a strongly right-skewed distribution. Table A3 reports the number and share of firms producing between one and ten or more four-digit products, together with the average number of employees and occupations within each product-scope category. More than half of all firms produce a single product across the sample period.

Table A3: Distribution of firms by number of products

Number of products	Firms	Share
1	2,090	0.533
2	839	0.214
3	350	0.089
4	178	0.045
5	108	0.028
6	71	0.018
7	48	0.012
8	32	0.008
9	21	0.005
≥ 10	185	0.047

Notes: This table contains the number of firms producing between 1 and more than 10 products in discrete categories. Shares are proportions of total firms. Product counts are computed across the full sample period from 2010 to 2019.

Table A4 reports the top ten products ranked by number of manufacturers and by average

sales, and Table A5 the ten most frequent bilateral co-production pairs. Both tables are discussed in Section 2.2.

Table A4: Top HS4 products by number of manufacturers and by mean sales

Ranked by number of manufacturers				Ranked by mean sales			
Rank	p	N_p	μ_p^{sales}	Rank	p	N_p	μ_p^{sales}
1	7308: steel structures	368	11,058	1	7308: steel structures	368	11,058
2	7326: iron/steel articles	340	6,205	2	2309: animal feed	47	8,393
3	3926: plastic articles	214	4,062	3	9403: furniture	207	8,130
4	7204: scrap iron/steel	214	152	4	9021: orthopedic devs	35	6,713
5	9403: furniture	207	8,130	5	7326: iron/steel articles	340	6,205
6	8479: machinery n.e.s.	205	3,830	6	4418: wood joinery	109	5,656
7	4911: printed matter	129	2,113	7	6810: cement articles	68	5,205
8	3923: plastic packaging	117	3,726	8	1905: bakery products	45	5,063
9	8428: lifting machinery	115	3,036	9	9031: measuring instr.	48	4,835
10	4418: wood joinery	109	5,656	10	8481: valves	60	4,827

Notes: This table reports the top 10 product categories (4 digits), denoted p, under two alternative rankings: by total number of manufacturers, denoted N_p, and by mean sales, denoted μ_p^{sales}. The latter is reported in thousands of DKK in 2015 values.

Table A6 quantifies how much of this cross-chapter co-production matters economically, as discussed in Section 2.2. Only 30.2 percent of firms manufacture across more than one two-digit HS chapter, yet this group accounts for 83.96 percent of total revenue and 80.1 percent of employment. Restricting to MPFs sharpens the pattern: 69.05 percent of MPFs produce across HS2 chapters, and this group generates 93.95 percent of MPF revenue and 92.6 percent of MPF employment. Firms that cross classification boundaries are therefore not a marginal subset but account for the large majority of manufacturing activity.

Table A5: Most common co-production pairs

Rank	ω_1	ω_2	N_{ω_1,ω_2}	$N_{\omega_1,\omega_2}/N$
1	7204: Iron and steel scrap	7308: Steel structures	81	0.0207
2	7204: Iron and steel scrap	7326: Steel articles	78	0.0199
3	7308: Steel structures	7326: Steel articles	64	0.0163
4	4901: Printed books	4911: Printed materials	43	0.0110
5	7326: Steel articles	7616: Aluminium articles	43	0.0110
6	9401: Seats	9403: Furniture	43	0.0110
7	3926: Plastic articles	7326: Steel articles	42	0.0107
8	8428: Lifting equipment	8431: Machinery parts	37	0.0094
9	4902: Newspapers	4911: Printed materials	34	0.0087
10	3926: Plastic articles	8480: Moulds and dies	31	0.0079

Notes: This table reports the most common product pairings. N_{ω_1,ω_2} denotes the number of firms producing both products at overlapping times during the sample period. N denotes the total number of unique firms present in the sample. Shares are proportions of all firms.

Table A6: Multi-HS2 production and economic importance

	Firms	Share of firms	Share of revenue	Share of employment
Panel A: All firms				
Single HS2	3,007	0.6977	0.1604	0.1990
Multi HS2	1,303	0.3023	0.8396	0.8010
Panel B: MPFs only				
Single HS2	584	0.3095	0.0605	0.0740
Multi HS2	1,303	0.6905	0.9395	0.9260

Notes: A firm is classified as "Multi HS2" if it manufactures products spanning more than one two-digit HS chapter across the sample period, and "Single HS2" otherwise. Panel A reports shares over the full firm sample (cf. Table A2); Panel B restricts to multi-product firms (MPFs, cf. Appendix B.1). "Share of revenue" and "Share of employment" report each row's share of total firm revenue and employment.

Figure A1 displays the kernel density of firms by organizational complexity, defined as the number of unique occupations observed across the sample period. The distribution ranges from one to more than fifty occupations, though very few firms register more than forty. The vast majority employ fewer than twenty, and the peak of the right-skewed distribution lies below ten.

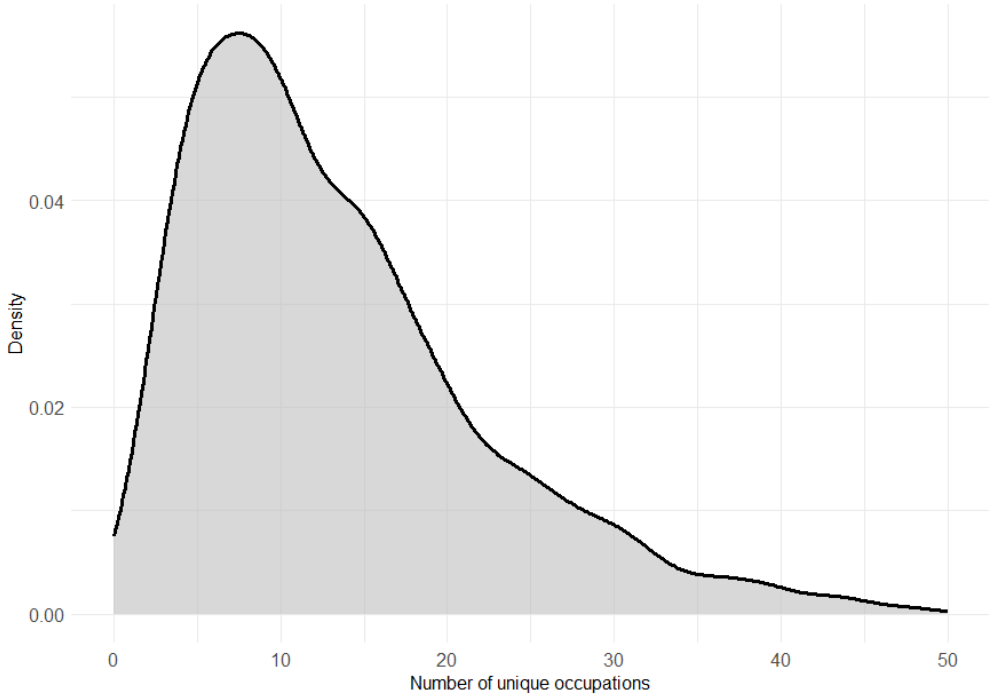


Figure A1: *This figure displays the kernel density of firms by organizational complexity, defined as the number of unique three-digit occupations a firm employs across the sample period.*

A.3 Worker and occupation summary statistics

This appendix reports summary statistics on workers and occupations (see Section 2.1). In total, 370,072 unique workers appear in the sample, of whom 29 percent are female. The average worker appears for 4.42 years and registers in 1.25 distinct three-digit occupations. Average working time is 1,412 hours per year, corresponding to 28.2 hours per week under a 250-working-day calendar and a five-day working week. Table A7 collects these statistics.

Some occupations are far more widely held than others; equivalently, some tasks are more commonly demanded by firms. Table A8 displays the ten most prevalent occupations across

Table A7: Worker-level summary statistics

	Gender	Workers	Years	Wage	Hours	Hourly wage	Occs (3-digit)
Mean	1.29	370,072	4.42	358,338.17	1,412.09	372.59	1.25

Notes: The table reports worker-level means. Wage is reported in levels, and hourly wage is computed as wage divided by hours. Occupation measures refer to the number of distinct occupations at the 3-digit level.

sample firms on the extensive margin. The counts are constructed from firms reporting any presence of a given occupation across the sample period and therefore do not capture the intensive margin, such as hours worked or wage-bill shares. Managerial occupations of several types (112, 132, 122, and 121, in ranked order of prevalence) are widespread. Within other one-digit categories, presence is more heterogeneous across individual occupations; among the most common are "411: General Office Clerks," "311: Physical/Engineering Technicians," and "332: Sales/Purchasing Agents & Brokers."

Table A8: Most prevalent occupations

DISCO	N firms	Share
112 Managing Directors & Chief Executives	2,684	0.757
132 Manufacturing, Construction & Distribution Managers	2,271	0.641
411 General Office Clerks	2,093	0.591
311 Physical & Engineering Technicians	2,092	0.590
332 Sales & Purchasing Agents/Brokers	1,833	0.517
331 Financial & Mathematical Associates	1,812	0.511
722 Blacksmiths & Toolmakers	1,778	0.502
122 Sales, Marketing & Development Managers	1,646	0.464
121 Business Services & Admin Managers	1,641	0.463
432 Materials & Transport Clerks	1,495	0.422

Notes:

A.4 Firm growth over the sample period

This appendix supplements Section 2.2. Figure A2 displays (a) growth of revenue, value added, employment, and yearly wages for the average firm (2010 = 100); (b) growth in the average number of products manufactured across HS aggregation levels (2010 = 100); and

(c) the underlying numeric growth of panel (b). Firms grow in revenue and value added over the period, alongside the product and employment growth documented in the main text.

The organizational counterpart of this growth is a mild increase in occupational scope: Figure A3 plots the average number of distinct occupations per firm by year, at one- through four-digit DISCO aggregation, which trends mildly upward over the sample period alongside the revenue, product, and employment growth in panels (a)–(b), though far less than proportionally, and the ranking across aggregation levels is stable throughout.

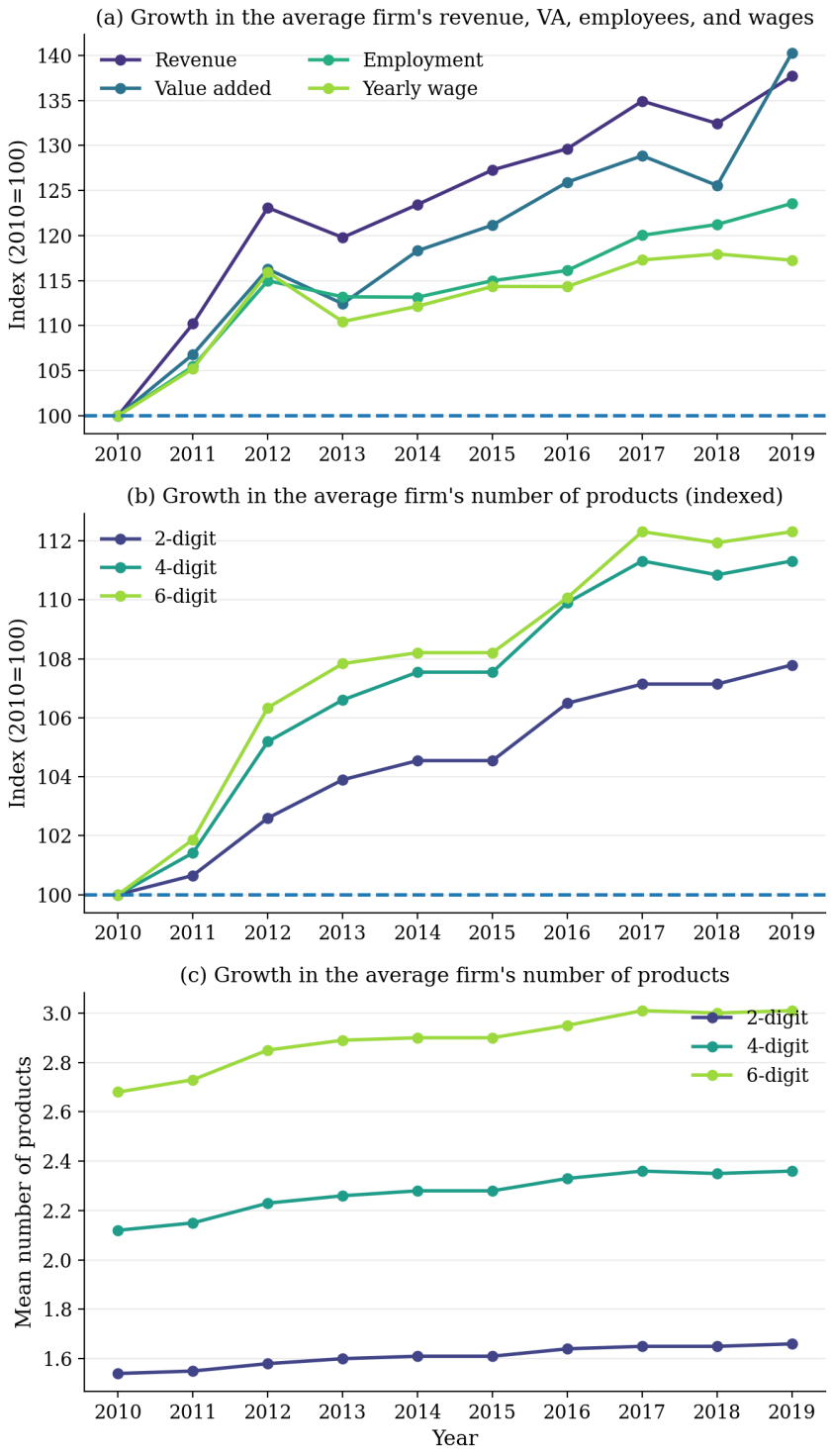


Figure A2: *This figure shows (a) the growth of revenue, value added (VA), employees, and yearly wages for the average firm (2010 = 100), (b) the growth in the average number of products manufactured (*$2010 = 100$*) across different aggregation levels of products, and (c) the numeric growth of panel (b).*

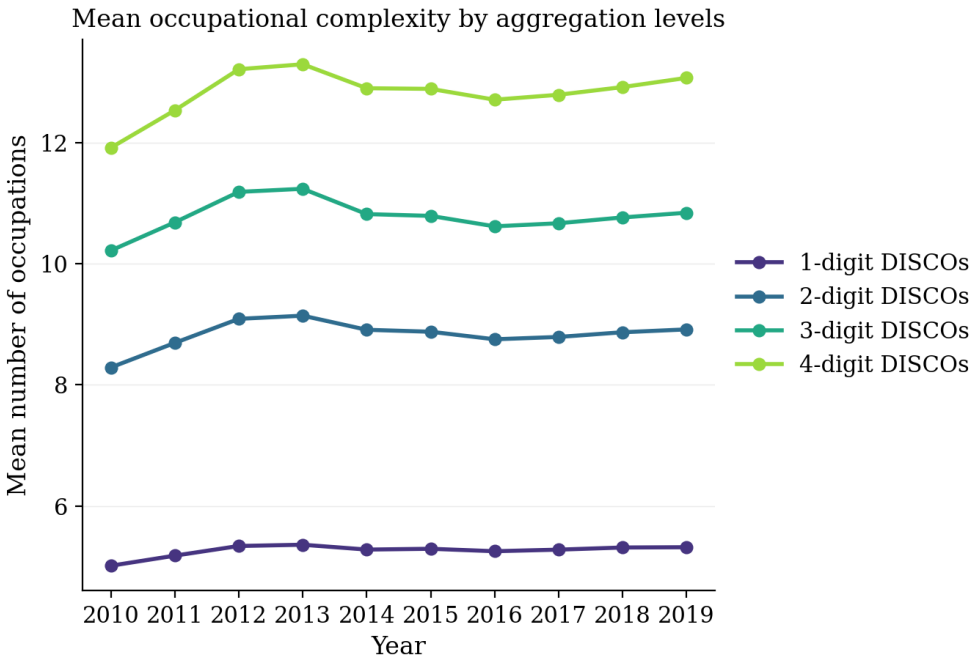


Figure A3: *This figure displays the average number of distinct occupations per firm, by year, separately at the one-, two-, three-, and four-digit DISCO aggregation levels.*

A.5 Occupational concentration and core-product predictiveness by scope

This appendix supplements Section 2.2. Table A9 reports mean occupation count and the occupational Herfindahl–Hirschman index by product scope, and Table A11 reports the numeric values underlying Figure 2.

Table A10 reports two decompositions of occupational structure by core product, split by scope bin. Panel A regresses each firm's occupation-level hours share on occupation fixed effects, with and without an interaction with the firm's core product; Panel B regresses firm-aggregate log occupation count and log HHI on core-product fixed effects alone.

Table A11 reports the average pairwise cosine similarity of firms' occupation vectors within core-product $\times$ scope-bin cells, for both hour-weighted and presence-only vectors, alongside a random-pairs benchmark drawn within the same scope bin.

Table A9: Occupation count and concentration by product scope

Scope bin	Firms	Mean occupations	Mean HHI
1 (SPF)	2,326	9.9	0.238
2	981	10.7	0.230
3	484	11.8	0.231
4–5	383	13.3	0.212
5–10	232	15.2	0.201
10+	129	19.6	0.149

Notes: Scope bins are each firm-year's distinct four-digit HS products manufactured that year, so a firm can appear in different bins in different years; the Firms column counts distinct firms observed in the bin in at least one year and therefore need not sum to the full firm count. Mean occupations is the average number of distinct three-digit DISCO codes a firm employs that year; mean HHI is the average, across firm-years in the bin, of the occupational Herfindahl–Hirschman index $\text{HHI}_{jt} = \sum_o \theta_{jot}^2$, where θ_{jot} is occupation o's share of firm j's total hours worked in year t (defined in Section 2.2).

Table A10: Core-product predictiveness of occupational structure, by scope

Panel A: Occupation-level hours share

Scope bin	Obs.	R^2: occupation FE	R^2: occupation × core-product FE
1 (SPF)	22,906	0.265	0.600
2	10,226	0.286	0.694
3	4,658	0.312	0.812
4–5	4,258	0.274	0.821
5–10	3,355	0.341	0.870
10+	3,537	0.268	0.890

Panel B: Firm-aggregate occupational structure

Scope bin	Obs.	R^2: log(occupations)	R^2: log(HHI)
1 (SPF)	13,325	0.233	0.296
2	4,708	0.327	0.384
3	1,869	0.482	0.554
4–5	1,492	0.492	0.534
5–10	903	0.589	0.689
10+	641	0.645	0.724

Notes: Both panels regress on core-product (four-digit HS) fixed effects alone, split by scope bin; standard errors are omitted since R^2 is the object of interest. Panel A uses each firm's occupation-level hours share θ_{jo} (Section 2.2), with hours and scope both pooled across the full sample period, and adds an occupation × core-product interaction in the second column. Panel B uses the firm-year panel underlying Fact 2, with scope measured each year; the two panels' scope bins are therefore not drawn from the same underlying variable.

Table A11: Between-firm occupational similarity within core-product × scope cells

		Hour-weighted		Presence only	
Scope bin	Cells	Same-core	Random	Same-core	Random
1 (SPF)	188	0.322	0.139	0.339	0.270
2	121	0.372	0.168	0.367	0.302
3	58	0.445	0.175	0.374	0.310
4–5	50	0.394	0.153	0.386	0.316
5–10	38	0.315	0.171	0.382	0.356
10+	28	0.436	0.206	0.489	0.419

Notes: Firms' occupation vectors pool hours across the full sample period (Section 2.2), and scope bins are likewise defined over the full period, not by year. "Same-core" is the average pairwise cosine similarity of these vectors within core-product (four-digit HS) × scope-bin cells with at least two firms, averaged across cells and weighted by cell size; "Random" is the average cosine similarity over 2,000 randomly drawn firm pairs within the same scope bin, unconditional on core product. Hour-weighted vectors use within-firm occupation hours shares; presence vectors are binary occupation indicators.

Appendix B SPF Definitions

B.1 Single and multi product firms

The strictest definition of an SPF is a firm that produces exclusively within a single product category, such that 100 percent of its sales are concentrated in that category. In practice, however, many firms generate at least a small fraction of sales outside their core product. As a result, the strict SPF definition covers only a limited set of products, leaving much of the product space uncovered. A tradeoff therefore arises: stricter definitions yield cleaner occupation–product vectors but sacrifice coverage, while looser definitions increase coverage at the cost of contamination from co-production. In Appendix B.2, I present an alternative approach to relaxing the strict SPF definition.

To fix notation, define the sales share of product ω in firm j's portfolio as:

$$\sigma_{j\omega} = \frac{R_{j\omega}}{R_j}, \tag{21}$$

where $R_{j\omega}$ denotes firm j's sales revenue from product ω and R_j its total sales revenue. This measure captures the relative importance of product ω for firm j. For ease of notation, I omit the time index t, although these shares are observed annually.

The canonical SPF definition requires concentration in a single product category. Formally:

$$\text{SPF}_j = \begin{cases} 1 & \text{if } \max_{\omega \in \Omega_j} \sigma_{j\omega} \geq \alpha, \\ 0 & \text{otherwise,} \end{cases} \tag{22}$$

where $\alpha \in (0, 1]$ is a threshold parameter. If $\text{SPF}_j = 0$, then $\text{MPF}_j = 1$ by construction, so every firm is classified as either SPF or MPF. Under the strict definition ($\alpha = 1$), only a small share of products are produced by SPFs. A natural relaxation is to lower α, which increases both the share of firms classified as SPFs and the coverage of products. This highlights the central tradeoff: stricter definitions reduce contamination but limit coverage, whereas looser definitions improve coverage but introduce noise from co-production.[23] Figure B1 illustrates

[23]The flipside of having more SPFs is fewer MPFs. Hence, if there is a subsequent aim to analyze MPFs,

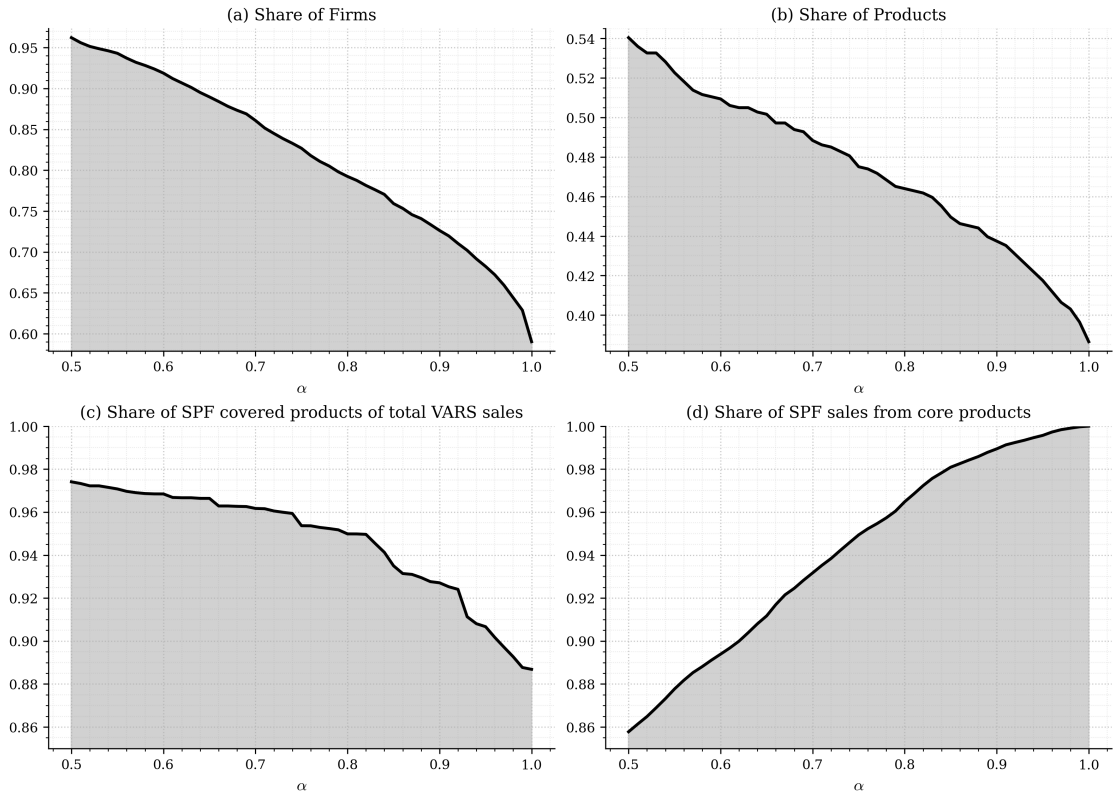


Figure B1: *This figure displays, as a function of the SPF threshold $\alpha \in (0.5, 1]$: (a) the share of firms classified as SPFs; (b) the share of products covered by SPF core products; (c) the share of total sales accounted for by SPF-produced products; and (d) the share of SPF sales accounted for by the core product alone.*

product coverage as a function of $\alpha \in (0.5, 1]$. Lower values of α expand SPF coverage of products (panel (b)), but at the expense of less pure occupation–product vectors, as co-produced activities receive greater weight (panel (d)). Panel (c) shows the share of total sales accounted for by SPF-produced products. Even under the strictest definition, these products account for a large share of economic activity, with nearly 90 percent of all sales coming from approximately 40 percent of products.

A key advantage of using SPF occupational vectors is that they rely on observed production data. The identifying assumption is that the observed occupational distribution reflects the tasks required to produce the firm's core product. Firms that produce exclusively within a single product category therefore provide the cleanest mapping from occupations to product-level task requirements. However, as shown in Figure B1, approximately 60 percent

this tradeoff may also affect sample size in later stages.

of firms produce entirely within one product code, yet these firms cover only 39 percent of products. While these products account for a large share of total economic activity (89 percent), these firms are relatively small and represent less than 35 percent of total sales (see Appendix B.1.1, panel (d)).

Even under the strict definition, there remains a case for relaxing the SPF criterion. The sharp decline in the share of SPFs when $\alpha > 0.95$ suggests that many firms produce a small fraction of output outside their core product. I argue that such firms are not representative of the canonical notion of multiproduct firms in the literature, but rather reflect minor deviations from otherwise concentrated production structures.

Appendix B.2 introduces an alternative approach to SPF definitions, where core-product thresholds are evaluated at different aggregation levels than those used to define the core product itself. For example, a firm may be classified as an SPF if it produces 100 percent within a single two-digit product category, while its core product is assigned at the four-digit level.

In practice, the preferred SPF definition balances coverage and purity: it should maximize product coverage while ensuring that occupational vectors remain representative of production tasks. I therefore adopt a definition where firms are classified as SPFs if at least 90 percent of their sales are concentrated in a single product category. Under this definition, SPFs account for 73 percent of firms and cover 44 percent of the product space through their core products. These products account for more than 92 percent of total sales, implying substantial coverage. Moreover, core products account for more than 98 percent of SPF sales, indicating that the resulting occupational vectors remain sufficiently clean.

B.1.1 Additional SPF metrics

Figure B2 complements Figure B1 with two further cuts of the same threshold sweep. Panel (b) recomputes panel (a) at the firm-year rather than firm level: because a firm's sales concentration varies from year to year, fewer firm-year observations than firms satisfy the SPF condition at any given α, so panel (b) sits below panel (a) throughout. Panels (c) and (d) shift from coverage to economic importance, reporting the share of total manufacturing sales accounted for by SPF core products and by SPFs overall; both decline in α for the

same reason as the coverage measures in Figure B1, but remain economically sizable even at the strictest threshold, underscoring that SPFs are not a marginal subset of production.

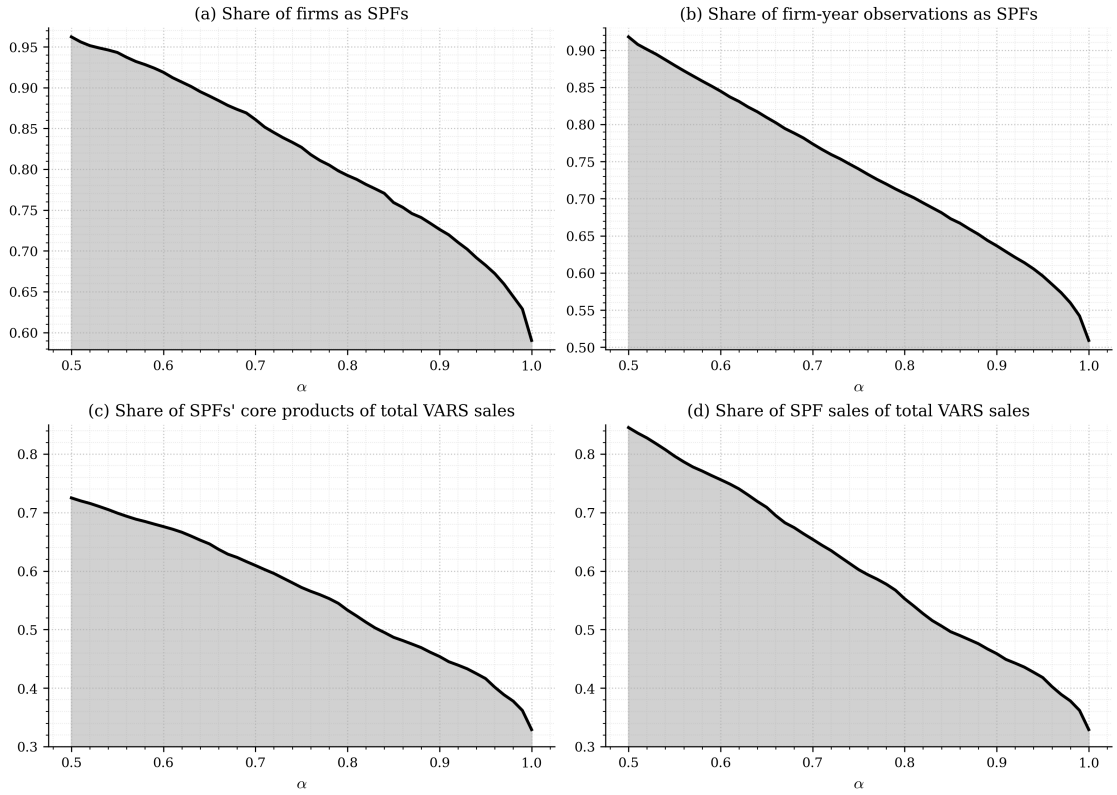


Figure B2: *This figure displays, as a function of the SPF threshold* $\alpha \in (0.5, 1]$*: (a) the share of firms classified as SPFs, repeated from Figure B1 for reference; (b) the share of firm-year observations classified as SPFs; (c) the share of total VARS sales accounted for by SPFs' core products; and (d) the share of total VARS sales accounted for by SPFs overall.*

B.2 Alternative SPF definition approach

A second approach to broadening the SPF concept is to classify firms at a coarser product aggregation level, which is one way of accepting a little co-production while ensuring cleanness. In this case, a firm is considered an SPF if most of its sales fall within a single product group at aggregation level k_1, even if these sales are spread across several sub-products within that group. Hence, SPF classifications are constructed at one level of HS aggregation following Equation 22, while assignment of SPF core product is carried out at another level, denoted k_2. For instance, k_1 in Equation 22 may refer to a 2-digit grouping, in which case all sales

within a single two-digit code are aggregated to test the SPF condition. Assignment of a firm's core product is then determined by the largest constituent product within the chosen group k_1^*. The product at level k_2 which has the highest sales share is then defined as the firm core-product, for which further analysis is conducted:

$$\text{If } \text{SPF}_j^{k_1} = 1 \quad \Rightarrow \quad \text{SPF}_{j\omega} = \begin{cases} 1 & \text{if } \omega \in k_1^* \text{ and } \sigma_{j\omega} = \max_{\omega' \in k_1^*} \sigma_{j\omega'}, \\ 0 & \text{otherwise,} \end{cases} \tag{23}$$

where $k_1^* \in HS^{(k_1)}$ is the core group for firm j, and $\sigma_{j\omega}$ denotes firm j's sales share of product ω as in Equation (22), here computed with products measured at k_2-digit resolution.

Most obviously, one would specify k_2 to be at a finer level than k_1: $k_2 > k_1$. One additional advantage of this method is that the relaxation at level k_1 should not lead to excessive contamination at level k_2 due to co-production of very distant products. As an example, consider a firm producing only within a two-digit code "grain products" (code 10), which will be classified as an SPF. Then, if the majority of sales is coming from one four-digit code "rice" (code 1006), this method will assign this firm as an SPF of product code 1006.

Table B1 displays the coverage of all possible levels of k_1 and k_2 in Equation 23, where $(k_1, k_2) \in \{2, 4, 6\} \times \{2, 4, 6\}$ are digit lengths. I report the share of firms defined as SPFs at this level, the products covered at aggregate k_1 level and detailed k_2 level, and the share of sales coming from SPFs. Importantly, the table also includes the share of all manufacturing sales from the products covered by SPF core products, and the share of SPF sales coming from core products. The former attempts to address the economic importance of the products covered by SPFs, while the latter addresses the cleanness of occupational vectors, as more clean measures of occupations needed to produce a product implies little production outside of the core product.

Table B1: Two-level SPF definition coverage

Aggr. (k_1^*)	Core (k_2)	Firms (share)	Products (aggr.)	Products (det.)	Sales (share)	Coverage k_2 (of all sales)	Cleanness k_2 (of SPF sales)
HS2	HS2	65.5%	77.1%	77.1%	41.9%	99.5%	100.0%
HS2	HS4	65.5%	77.1%	35.7%	41.9%	91.5%	93.9%
HS2	HS6	65.5%	77.1%	23.7%	41.9%	73.6%	89.4%
HS4	HS2	53.0%	31.4%	77.1%	29.4%	99.5%	100.0%
HS4	HS4	53.0%	31.4%	31.4%	29.4%	88.4%	100.0%
HS4	HS6	53.0%	31.4%	20.2%	29.4%	70.7%	94.7%
HS6	HS2	45.3%	17.7%	75.9%	21.8%	99.2%	100.0%
HS6	HS4	45.3%	17.7%	29.2%	21.8%	86.1%	100.0%
HS6	HS6	45.3%	17.7%	17.7%	21.8%	65.0%	100.0%

Notes: Percentages are rounded to one decimal place. "SPF" denotes single-product firms. Column k_1^* refers to HS chapters at the 2-, 4-, or 6-digit level; and column k_2 is the HS level used to classify a firm's core product following Equation 23. Column three, "Firms (share)" denotes the share of total firms defined as SPFs, column 4 "Products (aggr.)" displays the number of products at level k_1 covered by SPFs, column 5 "Products (det.)" the share of products covered by SPFs at level k_2, and column 6 "Sales (share)" the share of total sales registered by SPFs. The last two columns respectively show the coverage of SPF products; i.e. the share of total sales by all manufacturing firms that is covered by SPF products, and the cleanness of SPF sales; i.e. the share of SPF sales registered by their core-product.

Appendix C Bilateral Product Similarities and Occupational Flows

C.1 A numeric example of unweighted and weighted cosine similarities

This appendix contains calculations based on the example presented and illustrated in Section 3.1 in Figure 3. The numerical illustration is:

$$\mathcal{O} = \begin{pmatrix} 0.06 & 0.06 & 0.10 & 0.18 & 0.10 \\ 0.12 & 0.04 & 0.02 & 0.04 & 0.16 \\ 0.04 & 0.18 & 0.14 & 0.04 & 0.00 \end{pmatrix}$$

, where rows represent products, and columns represent occupations.

The corresponding cosine similarities would hence be:

$$\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle = 0.6827$$
$$\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_3) \rangle = 0.5997$$
$$\langle \mathcal{O}(\omega_2), \mathcal{O}(\omega_3) \rangle = 0.3343$$

such that $\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle > \langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_3) \rangle > \langle \mathcal{O}(\omega_2), \mathcal{O}(\omega_3) \rangle$.

Then, consider a weighting matrix Ψ where rows correspond to origin-occupations o and columns to destination-occupations o^*, which captures the estimated skill-transferability from an origin occupation o to a destination occupation o^*:

$$\Psi = \begin{pmatrix} 1 & 0.1 & 0 & 0 & 0 \\ 0.1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0.5 & 0 \\ 0 & 0 & 0.5 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

, where both rows and columns are occupations. Occupations 1 and 2 share skills or tasks to a small degree, and occupations 3 and 4 share skills to a larger degree. Occupation 5 does not share skills with any other occupation. Now, occupations are not only compared piecewise, but I compare all occupations across. Hence, for two products, all possible bilateral occupation pairs can increase the similarity, if there is presence and skill-sharing. The updated weighted similarities are now:

$$\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle = 0.6637$$
$$\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_3) \rangle = 0.7207$$
$$\langle \mathcal{O}(\omega_2), \mathcal{O}(\omega_3) \rangle = 0.4125$$

, and hence the new ranking of similarities is $\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_3) \rangle > \langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle > \langle \mathcal{O}(\omega_2), \mathcal{O}(\omega_3) \rangle$. This example hence serves to illustrate the importance of considering substitutability or skill-sharing between detailed occupation pairs, as products ω_1 and ω_3 now share more occupational overlap than ω_1 and ω_2. In general, the occupational similarity magnitude is hence determined by the relative importance and the size of the estimated skill-sharing.

To formally write out the calculations, consider first the unweighted cosine similarity between ω_1 and ω_2:

$$\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle = \frac{\mathcal{O}(\omega_1)^\top \mathcal{O}(\omega_2)}{\|\mathcal{O}(\omega_1)\| \cdot \|\mathcal{O}(\omega_2)\|}.$$

The inner product is:

$$\begin{aligned} \mathcal{O}(\omega_1)^\top \mathcal{O}(\omega_2) &= 0.06 \cdot 0.12 + 0.06 \cdot 0.04 + 0.10 \cdot 0.02 + 0.18 \cdot 0.04 + 0.10 \cdot 0.16 \\ &= 0.0072 + 0.0024 + 0.0020 + 0.0072 + 0.0160 \\ &= 0.0348. \end{aligned}$$

The magnitudes are:

$$\begin{aligned} \|\mathcal{O}(\omega_1)\| &= \sqrt{0.06^2 + 0.06^2 + 0.10^2 + 0.18^2 + 0.10^2} = \sqrt{0.0596}, \\ \|\mathcal{O}(\omega_2)\| &= \sqrt{0.12^2 + 0.04^2 + 0.02^2 + 0.04^2 + 0.16^2} = \sqrt{0.0436}. \end{aligned}$$

Thus,

$$\begin{aligned} \langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle &= \frac{0.0348}{\sqrt{0.0596}\sqrt{0.0436}} \\ &= 0.6827. \end{aligned}$$

Next, consider the weighted cosine similarity:

$$\Phi(\omega_1, \omega_2) = \frac{\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle_\Psi}{\sqrt{\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_1) \rangle_\Psi} \sqrt{\langle \mathcal{O}(\omega_2), \mathcal{O}(\omega_2) \rangle_\Psi}}$$

The numerator is:

$$\begin{aligned}\mathcal{O}(\omega_1)^\top \Psi \mathcal{O}(\omega_2) &= 0.06(1 \cdot 0.12 + 0.1 \cdot 0.04) \\ &\quad + 0.06(0.1 \cdot 0.12 + 1 \cdot 0.04) \\ &\quad + 0.10(1 \cdot 0.02 + 0.5 \cdot 0.04) \\ &\quad + 0.18(0.5 \cdot 0.02 + 1 \cdot 0.04) \\ &\quad + 0.10(1 \cdot 0.16) \\ &= 0.0368.\end{aligned}$$

The weighted magnitudes are:

$$\begin{aligned}\mathcal{O}(\omega_1)^\top \Psi \mathcal{O}(\omega_1) &= 0.0796, \\ \mathcal{O}(\omega_2)^\top \Psi \mathcal{O}(\omega_2) &= 0.0452.\end{aligned}$$

Hence,

$$\begin{aligned}\langle \mathcal{O}(\omega_1), \mathcal{O}(\omega_2) \rangle_\Psi &= \frac{0.0368}{\sqrt{0.0796}\sqrt{0.0452}} \\ &= 0.6637.\end{aligned}$$

Which is calculated from the following R-code:

```
# Function to calculate (unweighted) cosine similarity
fn_cosine_sim <- function(vec1, vec2) {

  inner_prod <- sum(vec1 * vec2)
  magnitude_v1 <- sqrt(sum(vec1^2))
  magnitude_v2 <- sqrt(sum(vec2^2))

  sim <- inner_prod / (magnitude_v1 * magnitude_v2)

  return(sim)
}

```

```
# Function to calculate weighted cosine similarity
fn_weighted_cosine_sim <- function(vec1, vec2, mat_weights) {

  # numerator
  inner_prod <- t(vec1) %*% mat_weights %*% vec2

  # denominator terms
  magnitude_v1 <- sqrt(t(vec1) %*% mat_weights %*% vec1)
  magnitude_v2 <- sqrt(t(vec2) %*% mat_weights %*% vec2)

  # output
  sim <- inner_prod / (magnitude_v1 * magnitude_v2)

  return(as.numeric(sim))
}

# Example vectors: shares of labor input across occupations
vO1 = c(0.06, 0.06, 0.10, 0.18, 0.10)
vO2 = c(0.12, 0.04, 0.02, 0.04, 0.16)
vO3 = c(0.04, 0.18, 0.14, 0.04, 0.00)

# Example weighting matrix (occupation similarity)
mWeights <- matrix(c(
  1,   0.1, 0,   0,   0,
  0.1, 1,   0,   0,   0,
  0,   0,   1,   0.5, 0,
  0,   0,   0.5, 1,   0,
  0,   0,   0,   0,   1
), nrow = 5, byrow = TRUE)

# --- Unweighted cosine similarities ---
fn_cosine_sim(vO1, vO2) # 0.6827
fn_cosine_sim(vO1, vO3) # 0.5997
fn_cosine_sim(vO2, vO3) # 0.3343

# --- Weighted cosine similarities ---
```

```
49 fn_weighted_cosine_sim(v01, v02, mWeights) # 0.6637
50 fn_weighted_cosine_sim(v01, v03, mWeights) # 0.7207
51 fn_weighted_cosine_sim(v02, v03, mWeights) # 0.4125
```

C.2 Occupational flows: mobility rates, asymmetry, and sparsity

This appendix documents the properties of the observed worker flow matrix $\mathcal{F}_{o,o^*}$ used to construct the skill-transferability matrix Ψ in Section 3.2, and discusses the resulting estimates in more detail.

Occupational mobility rates. Groes et al. (2015) report that close to one fifth of workers in Denmark change occupations within a year, which is slightly higher than the figures reported by Humlum (2021a; 2021b). In the present data, a little more than 10 percent of workers shift occupations in a given year after 2011. Similar magnitudes are found for the United States by Moscarini and Thomsson (2007) and Kambourov and Manovskii (2008), and these are more in line with the magnitude of job switches in my sample. Table C.2 reports that between 115 thousand and 153 thousand workers switch occupations each year, corresponding to between 7.8 and 10.3 percent of the worker sample.

Table C1: Job switchers by year

Year	Workers	Switchers	Share
2011	1,490,079	141,024	0.0946
2012	1,489,660	115,478	0.0775
2013	1,491,990	117,488	0.0787
2014	1,491,852	153,843	0.1031
2015	1,514,741	127,323	0.0841
2016	1,496,945	124,446	0.0831
2017	1,479,878	123,971	0.0838
2018	1,488,514	121,239	0.0814
2019	1,522,072	122,917	0.0808

Notes: Number of workers, number of switching to a new occupation, and the resulting switch share by year.

Within- versus between-firm transitions. I do not distinguish between job switchers and workers who change occupational codes within ongoing employment relationships. Groes et al. (2015) show that occupational transitions are more frequent among job switchers, but a substantial share of transitions also occur within firms, suggesting that internal labor markets play an important role in generating occupational flows. In the present data, approximately

61 percent of occupational changes occur in relation to a job switch.[24] Around half of the occupation changes move upward across the numerical DISCO categories, and the remainder move downward; Table C.2 reports the full breakdown.

Table C2: Job switches by hierarchical direction

	Up hierarchy		
	No	Yes	Total
Employer switch: No	4,637	4,290	8,927
Employer switch: Yes	7,014	6,979	13,993
Total	11,651	11,269	22,920

Notes: This table reports the number of worker flows between occupations, split into flows within and between employers, and whether the flow is up or down the numerical hierarchy of the occupation classifications (see https://www.dst.dk/da/Statistik/dokumentation/nomenklaturer/disco-loen).

Asymmetry of bilateral flows. Within particular occupation codes, there is asymmetry between inflows and outflows of workers, which supports the appropriateness of an asymmetric specification of Ψ_{o,o^*}. Figures C1–C4 illustrate observed worker in- and outflows for selected occupations.

Sparsity of bilateral flows. At the level of bilateral occupation pairs, observed flows are sparse. Approximately half of all occupation pairs exhibit zero flows, while nearly ten percent of pairs involve only a single worker transition. Figure C5 presents a binned histogram of bilateral occupation pairs by flow count. These patterns support the use of a model that explicitly accounts for zero-inflation. The ZINB specification is particularly suitable, as it jointly models (i) the probability that a positive flow occurs and (ii) the size of the flow, capturing both the extensive and intensive margins of skill transferability.

ZINB estimation results. Estimation results are presented in Table C3: higher average wages, wage growth rates, and employment growth in a destination occupation are associated

[24] An occupation change for a job switcher is defined by a simultaneous change in employer ID and occupation code for the worker from $t-1$ to t.

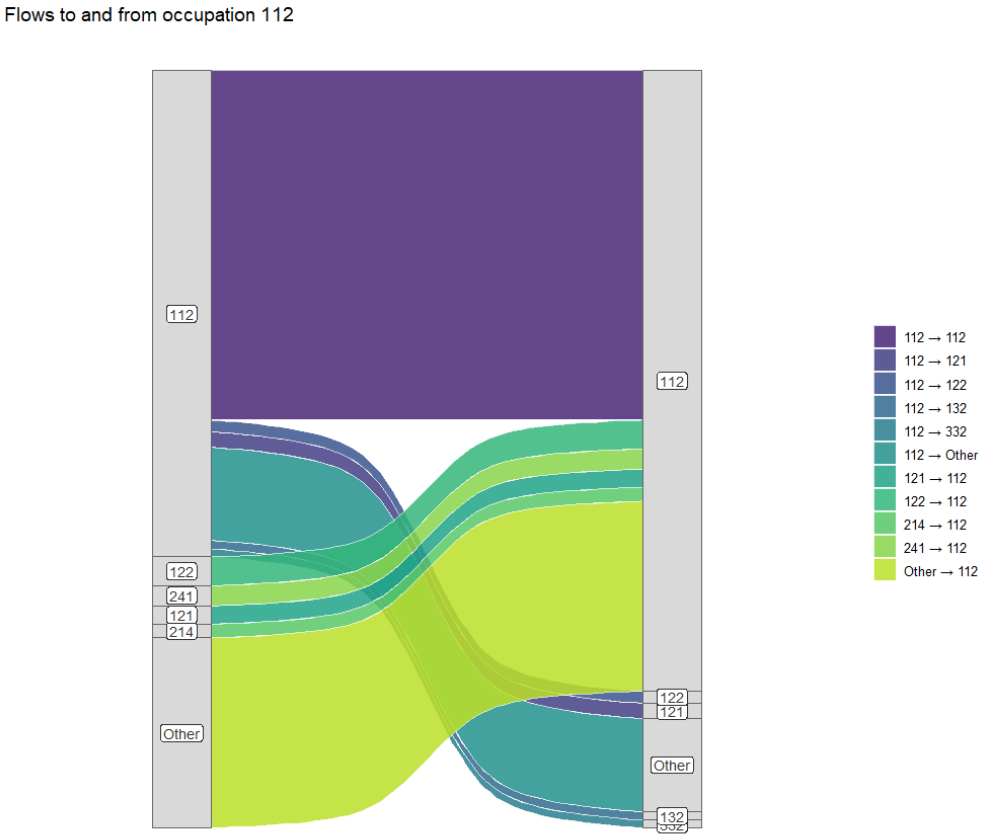


Figure C1: *This figure displays the in- and outflows of occupation 112: Admin managers.*

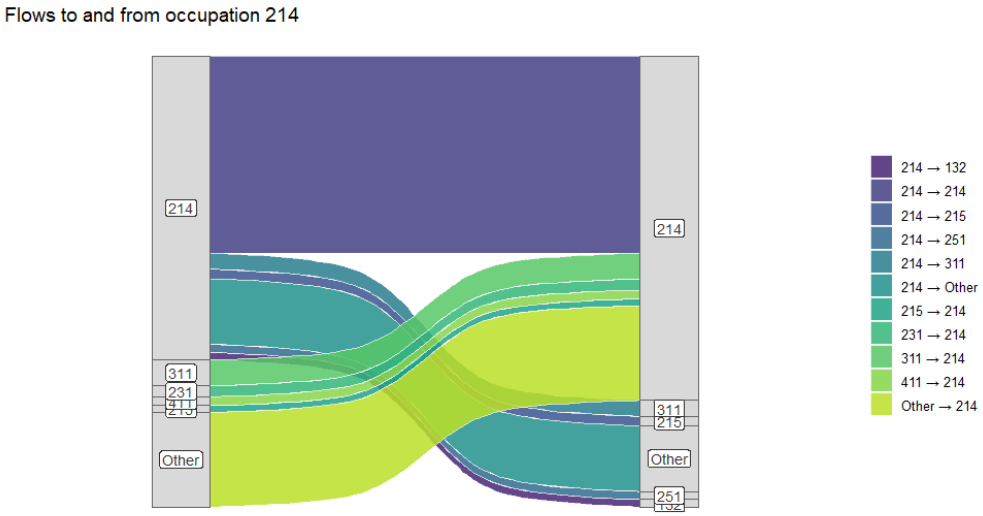


Figure C2: *This figure displays the in- and outflows of occupation 214: Engineers.*

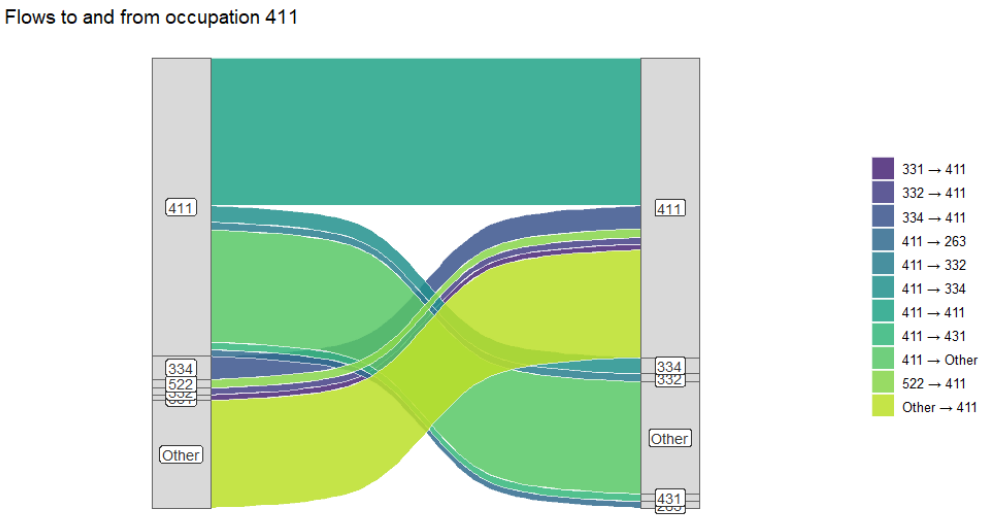


Figure C3: *This figure displays the in- and outflows of occupation 411: Office clerks.*

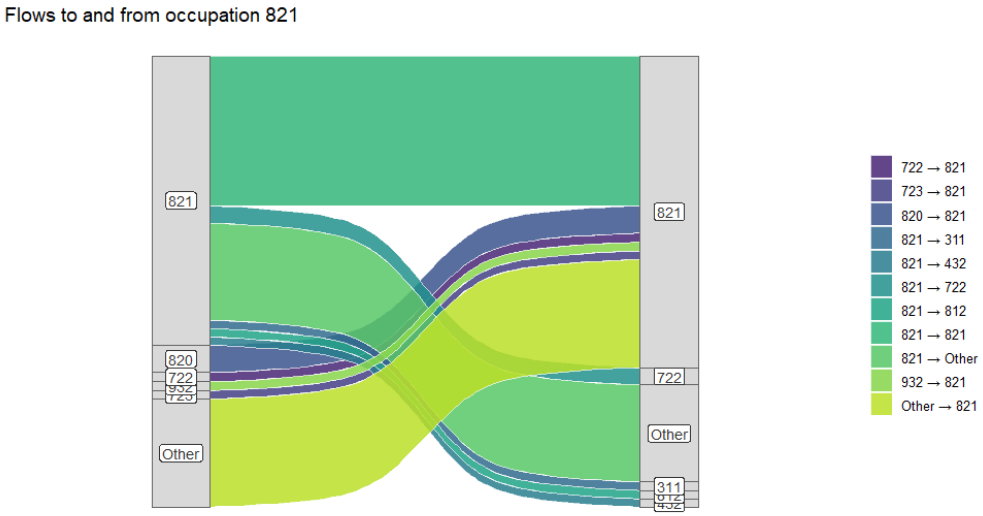


Figure C4: *This figure displays the in- and outflows of occupation 821: Assembly work.*

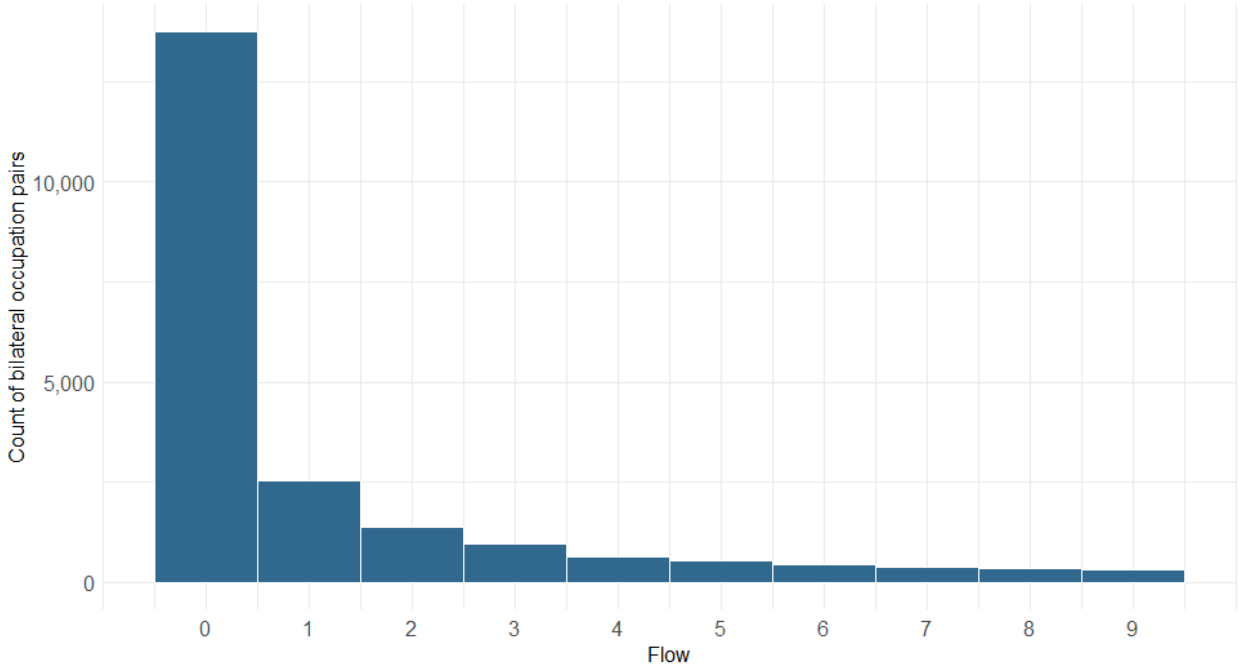


Figure C5: *This figure displays the distribution of bilateral occupation pairs by observed flow count $\mathcal{F}_{o,o^*}$, binned.*

with larger inflows, and occupations with larger employment levels tend to receive more workers. All occupations with sufficiently large employment are estimated to experience positive inflows over the sample period.

High skill-transferability pairs. The occupation pairs with the highest estimated skill transferability are reported in Table C4. Occupations within the same main categories (see Section 2.1) often exhibit high skill relatedness. For instance, "232: Vocational Education Teachers" has a high estimated skill relatedness with "233: Secondary Education Teachers"; "713: Painters/Building Cleaners" with "712: Building Finishers"; and "721: Metal Workers/Welders" with "722: Blacksmiths/Toolmakers". However, a substantial number of high-Ψ pairs span one-digit groups, indicating that skill transferability is not simply a relabeling of the DISCO hierarchy. For instance, "512: Cooks" have a high excess worker flow to "343: Arts/Cultural/Culinary Associates" (which includes "3434: Kitchen Chef") and to "941: Food Preparation Assistants", as both an origin and destination occupation. "541: Protective Service Workers" (which includes "5419: Ambulance Rescuers" among other relevant four-digit occupations) relate highly to "325: Other Health Associates" and

Table C3: Zero-inflated negative binomial model of bilateral occupation flows

	Coefficient
Count model	
Intercept	0.94^{***}
	(0.02)
total employment$_i$	1.05^{***}
	(0.03)
total employment$_{i*}$	0.91^{***}
	(0.02)
avg. wage$_i$	0.06^{**}
	(0.02)
avg. wage$_{i*}$	0.16^{***}
	(0.02)
employment growth$_i$	-0.07^{***}
	(0.02)
employment growth$_{i*}$	0.09^{***}
	(0.03)
wage growth$_i$	-0.22^{***}
	(0.04)
wage growth$_{i*}$	0.25^{**}
	(0.09)
Log(theta)	-1.53^{***}
	(0.02)
Zero model	
Intercept	-20.63^{***}
	(1.96)
total employment$_i$	-0.65^{***}
	(0.05)
total employment$_{i*}$	-37.44^{***}
	(3.31)
avg. wage$_i$	-0.15^{**}
	(0.05)
avg. wage$_{i*}$	-0.39^{***}
	(0.06)
employment growth$_i$	-0.06
	(0.07)
employment growth$_{i*}$	-0.18^{***}
	(0.05)
wage growth$_i$	-0.02
	(0.12)
wage growth$_{i*}$	0.21^{***}
	(0.06)
AIC	66755.11
Log Likelihood	-33358.56
Number of Observations	$23,244$

$^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$

"224: Paramedical Practitioners" (the latter as destination).

Generic occupations. In general, some occupations entail more transferable skills than others. Ranking occupations o by their mean skill relatedness $\bar{\Psi}_{o,\cdot}$ to all possible destination occupations o^*, "224: Paramedical Practitioners" has the highest mean skill relatedness but

Table C4: Top occupation pairs by skill sharing.

Origin o	Destination o^*	$\mathcal{F}_{o,o^*}$	Ψ_{o,o^*}
512: Cooks	343: Arts/Cultural/Culinary Assoc.	1219	152.8016
232: Vocational Educ. Teachers	233: Secondary Educ. Teachers	2168	99.7396
512: Cooks	941: Food Prep. Assistants	1803	98.6495
713: Painters/Building Cleaners	712: Building Finishers	1908	85.8258
541: Protective Service Workers	325: Other Health Associates	2009	82.7434
224: Paramedical Practitioners	541: Protective Service Workers	91	76.5060
252: Database/Network Prof.	351: ICT Ops/User Support	945	74.9422
612: Animal Producers	324: Veterinary Techs/Assistants	47	68.4551
721: Metal Workers/Welders	722: Blacksmiths/Toolmakers	2061	67.9408
941: Food Prep. Assistants	512: Cooks	1615	63.4599

Notes: This table contains the top 10 bilateral occupation pairs in terms of estimated excess skill-transferability from any origin occupation o to a destination occupation o^*. For each bilateral pair, the table contains the observed flow $\mathcal{F}_{o,o^*}$ and the excess flow estimation Ψ_{o,o^*} following Equation 3, which is controlled for occupation-specific size and wage characteristics potentially driving flows. The expected flow is estimated using a ZINB model accounting for a large share of observed 0s. All text labels of occupations are generated using Anthropic's Claude from author provided text descriptions in Danish, prompted to translate and shorten occupational titles for readability purposes.

relatively few observed destinations. Occupations "512: Cooks" and "343: Arts/Cultural/-Culinary Associates" are also highly generic in skills and have more observed destination occupations. Particularly relevant for the sample used, "821: Assemblers" is highly generic, as are both "818: Other Plant/Machine Operators" and "721: Metal Workers/Welders", among other manufacturing-oriented jobs.

C.3 Overview of product similarities

Top 30 occupationally similar products The pairwise similarities underlying Figure 4 are dominated, at the very top of the distribution, by pairs of products that plausibly share production processes or inputs rather than belonging to the same conventional product category. Table C5 lists the 30 most similar HS4 product pairs. Several clusters emerge: metal-working intermediates and parts (ferrous scrap, steel pipes, machine tool parts, moulding boxes), food products sharing meat- or fish-based inputs (sausages, prepared fish, meat), and machinery components (grain-processing, woodworking, and lifting/handling equipment). This clustering is consistent with the measure picking up shared

task content rather than merely replicating the HS classification's own product groupings, since several of the highest-ranked pairs sit in different two-digit HS chapters.

Table C5: Most closely related HS product pairs

Rank	Product 1	Product 2	Similarity
1	7325: Cast iron articles	7419: Other copper articles	0.988
2	8207: Interchangeable tools	8480: Moulding boxes	0.986
3	8466: Machine tool parts	8480: Moulding boxes	0.981
4	1601: Sausages	1604: Prepared fish	0.980
5	7304: Steel pipes (seamless)	8466: Machine tool parts	0.979
6	8207: Interchangeable tools	8466: Machine tool parts	0.977
7	8431: Machinery parts	8437: Grain processing machinery	0.977
8	7326: Other iron/steel articles	8707: Vehicle bodies	0.976
9	7204: Ferrous scrap	7304: Steel pipes (seamless)	0.974
10	0201: Meat	1601: Sausages	0.973
11	8422: Dishwashing/packing machinery	8428: Lifting/handling equipment	0.973
12	8431: Machinery parts	8465: Woodworking machines	0.973
13	7204: Ferrous scrap	8707: Vehicle bodies	0.973
14	8465: Woodworking machines	8707: Vehicle bodies	0.972
15	7326: Other iron/steel articles	8437: Grain processing machinery	0.970
16	7326: Other iron/steel articles	8431: Machinery parts	0.970
17	7304: Steel pipes (seamless)	8480: Moulding boxes	0.970
18	8431: Machinery parts	8707: Vehicle bodies	0.968
19	0304: Fish fillets	1601: Sausages	0.967
20	3923: Plastic packaging	3926: Other plastic articles	0.966
21	1601: Sausages	1704: Sugar confectionery	0.965
22	4911: Other printed matter	8442: Printing machinery	0.964
23	3920: Plastic sheets	4819: Paper packaging	0.964
24	0201: Meat	1604: Prepared fish	0.963
25	7204: Ferrous scrap	8480: Moulding boxes	0.963
26	4821: Paper labels	4901: Printed books	0.963
27	7204: Ferrous scrap	8465: Woodworking machines	0.963
28	7204: Ferrous scrap	8207: Interchangeable tools	0.963
29	7309: Steel tanks	8609: Containers	0.963
30	7304: Steel pipes (seamless)	8207: Interchangeable tools	0.962

Notes: The table reports the most closely related HS4 product pairs based on a similarity measure. Higher values indicate greater similarity in production or input structure.

Distribution of bilateral product similarities Figure C6 plots the full distribution of pairwise similarities across the GDPR-approved product space depicted in Figure 4. The distribution is sparse and right-skewed: most product pairs share little task content, with density concentrated around $\Phi \approx 0.2$, and the high-similarity pairs in Table C5 occupy the thin right tail.

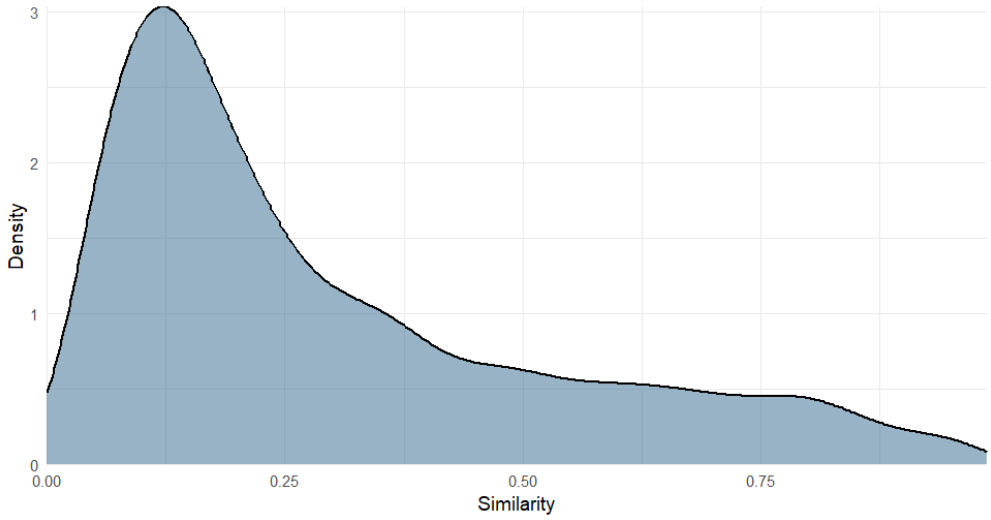


Figure C6: *This figure displays the distribution of similarities of bilateral product pairs across the GDPR-approved product space also depicted in Figure 4, on the basis of Equation 2. The distribution is right-skewed; product pairs most frequently have an occupational similarity around* 0.2.

Appendix D Theoretical Framework for Scope Adjustments

This appendix collects the primitives of Eckel and Neary (2010) that the extension in Section 4 takes as given, restated in the notation used throughout this paper. The purpose is threefold: to make explicit where the viability condition in Equation (5) originates; to clarify how the abstract cost-ordering index of the original model maps into the observable similarity measure $\Phi(\bar{\omega}_j, \omega)$; and to identify which components of the net surplus $s_{j\omega}$ in Equation (6) are absorbed by the fixed effects in Section 5. Appendices D.1–D.3 contain no new results; the reindexing in Appendix D.4 and the threshold derivation in Appendix D.5

operationalize the extension, and Appendices D.6–D.7 discuss its empirical content.

D.1 Preferences and demand

Consumers have two-tier preferences. The upper tier is additively separable across a continuum of industries indexed by $z \in [0,1]$, $U = \int_0^1 u\{z\}\, dz$, so that industries do not interact through the upper tier. Within an industry, a mass N of horizontally differentiated varieties enters a quadratic sub-utility function,

$$u\{z\} = a \int_0^N q(\omega)\, d\omega - \frac{1}{2} b \left[(1-e) \int_0^N q(\omega)^2 d\omega + e \left(\int_0^N q(\omega)\, d\omega \right)^2 \right], \tag{24}$$

which combines the continuum-quadratic treatment of symmetric horizontal differentiation in **?** with the omission of an outside good as in **?**.The parameters a, b, and e are non-negative and common across consumers. Here a is the maximum willingness to pay and $e \in (0,1)$ is an inverse index of product differentiation: as $e \to 1$ varieties become perfect substitutes and only aggregate output matters for demand, while $e \to 0$ delivers independent demands. Both limits are excluded, since the mechanism of interest requires varieties to compete with one another without being interchangeable.

Maximizing Equation (24) subject to the budget constraint $\int_0^1 \int_0^N p(\omega) q(\omega)\, d\omega\, dz \leq I$ yields the individual inverse demand

$$\lambda\, p(\omega) = a - b \left[(1-e) q(\omega) + e \int_0^N q(\omega')\, d\omega' \right], \tag{25}$$

where λ is the multiplier on the budget constraint and therefore the marginal utility of income. Aggregating across L consumers in each of k countries with integrated goods markets and a common price for each variety, market demand is $x(\omega) = kL\, q(\omega)$, and Equation (25) becomes the inverse market demand curve

$$p(\omega) = a' - b' \left[(1-e)\, x(\omega) + eY \right], \qquad a' \equiv \frac{a}{\lambda}, \quad b' \equiv \frac{b}{\lambda k L}, \quad Y \equiv \int_0^N x(\omega')\, d\omega'. \tag{26}$$

This defines the objects a', b', e, and Y that appear in Equation (5). Two features matter for

the empirical work. First, the demand slope b' falls with the size of the integrated market kL, so trade integration operates in part by flattening demand. Second, although a' and b' are endogenous in general equilibrium through λ, the continuum of industries means each firm treats them as exogenous: firms are large in their own market but negligible economy-wide. Following the original treatment I normalize $\lambda = 1$, so all nominal magnitudes are expressed relative to the marginal utility of income.

D.2 Technology and profits

Labor is the only factor of production and the labor market is economy-wide and competitive, so the marginal cost of a variety separates into a wage and a technological component, $c_j(\omega) = W\gamma_j(\omega)$, where $\gamma_j(\omega)$ is the unit labor requirement. Marginal cost is constant in the quantity produced but varies across varieties: it is lowest at the firm's core product $\bar{\omega}_j$ and rises as the firm moves away from it. Crucially, flexible manufacturing means that adding a product raises the marginal cost of the *new* product only, leaving incumbent products' costs unchanged. This distinguishes the framework from models in which scope expansion raises costs across the entire portfolio, and it is what makes the pairwise object $\Phi(\bar{\omega}_j, \omega)$ the natural sufficient statistic for the adoption decision. Writing Ω_j for firm j's portfolio, profits are

$$\Pi_j = \int_{\Omega_j} \left[p_j(\omega) - c_j(\omega)\right] x_j(\omega)\, d\omega - F, \tag{27}$$

with the fixed cost F independent of both scale and scope.

D.3 Optimal scale, cannibalization, and optimal scope

Firms play a single-stage Cournot game, choosing quantities and scope simultaneously and taking rivals' choices as given. The first-order condition for the scale of an individual product is

$$p_j(\omega) - c_j(\omega) = b' \left[(1-e)\, x_j(\omega) + eX_j\right], \qquad X_j \equiv \int_{\Omega_j} x_j(\omega')\, d\omega', \tag{28}$$

so the optimal margin is proportional to a weighted average of own output and the firm's total output. Eliminating $p_j(\omega)$ between Equations (26) and (28) gives

$$x_j(\omega) = \frac{a' - c_j(\omega) - b'e\,(X_j + Y)}{2b'(1-e)}. \tag{29}$$

Industry output Y depresses own output through the usual Cournot competition channel, while the firm's own aggregate output X_j enters separately: expanding any one product lowers the price consumers will pay for every other product in the portfolio, giving the MPF an incentive to restrict output beyond the own-price effect. This is the cannibalization effect, and it is the reason X_j and Y enter Equation (5) additively as $X_j + Y$. Solving for prices and margins gives

$$p_j(\omega) = \tfrac{1}{2}\left[a' + c_j(\omega) - b'e\,(Y - X_j)\right], \qquad p_j(\omega) - c_j(\omega) = \tfrac{1}{2}\left[a' - c_j(\omega) - b'e\,(Y - X_j)\right], \tag{30}$$

so that exactly half of any cost disadvantage is passed to consumers and half is absorbed in a lower margin, a consequence of demand linearity. Products further from the core therefore carry both higher prices and lower margins.

The scope margin follows from differentiating Equation (27) with respect to the extent of the product range. The resulting condition, $\left[p_j(\underline{\omega}_j) - c_j(\underline{\omega}_j)\right] x_j(\underline{\omega}_j) = 0$, cannot be satisfied by a zero margin, since Equation (28) makes the margin strictly positive for any produced variety. It must therefore be satisfied by driving output of the marginal product to zero, $x_j(\underline{\omega}_j) = 0$, which by Equation (29) is equivalent to

$$c_j(\underline{\omega}_j) = a' - b'e\,(X_j + Y)\,. \tag{31}$$

The firm extends its range outward from the core until the marginal cost of the last product equals marginal revenue evaluated at zero output. For any candidate product, viability then requires $x_j(\omega) \geq 0$ in Equation (29), which is exactly Equation (5) once $c_j(\omega)$ is replaced by the parameterization in Equation (4). One difference in interpretation is worth recording: with a continuum of products the marginal product is produced in vanishing quantity, whereas the empirical product space is discrete (HS4), so the marginal *observed*

product is produced in strictly positive quantity and Equation (5) generally holds with slack. It is the weak inequality, not the equality in Equation (31), that carries to the data.

D.4 From the cost ordering to task similarity

In the original framework, products are indexed by their position in the firm's cost ordering: the index is constructed so that the unit labor requirement is increasing in distance from the core, and it therefore has no counterpart in observable data. The extension in Section 4 replaces this ordering device with a measurable one by writing the technological component as

$$\gamma_j(\omega) = \gamma_\omega^0 + \kappa \left(1 - \Phi(\bar{\omega}_j, \omega)\right)^\eta . \tag{32}$$

Three points follow. First, Equation (32) nests the linear illustration used in the original paper: setting $\eta = 1$ and identifying the ordering index with task distance, $1 - \Phi$, recovers a unit labor requirement that is linear in distance from the core with slope κ. The curvature parameter η generalizes this, and is the object identified by the percentile-binned specification in Section 5. Second, the ordering of candidate products is now firm-specific in a substantive rather than a labelling sense: two firms with different core products rank the same candidate differently, because Φ is evaluated at different $\bar{\omega}_j$. This is what generates cross-firm variation in the viability of a given product at a given point in time. Third, γ_ω^0 is indexed by the product alone, not by the firm. Its role is to allow products to differ in baseline task intensity for reasons unrelated to any particular firm's competences; when it is common across candidates, the viable set is an interval in similarity space, as noted in Section 4.

D.5 Similarity threshold for production

This appendix derives the similarity threshold $\Phi^*(\bar{\omega}_j, \omega)$ stated in Section 4, by solving the viability condition of Equation (5) for Φ. I start from Equation 5, and consider the case where the inequality is binding:

$$a' - W \cdot \left[\gamma_\omega^0 + \kappa(1 - \Phi(\bar{\omega}_j, \omega))^\eta\right] - b' e \left[X_j + Y\right] = 0$$

Then, I add the cost function to both sides:

$$W \cdot \left[\gamma_\omega^0 + \kappa(1 - \Phi(\bar{\omega}_j, \omega))^\eta\right] = a' - b'e\left[X_j + Y\right]$$

And divide by W:

$$\gamma_\omega^0 + \kappa(1 - \Phi(\bar{\omega}_j, \omega))^\eta = \frac{a' - b'e\left[X_j + Y\right]}{W}$$

And subtract γ_ω^0:

$$\kappa(1 - \Phi(\bar{\omega}_j, \omega))^\eta = \frac{a' - b'e\left[X_j + Y\right]}{W} - \gamma_\omega^0$$

Next, I divide by κ, and multiply W to γ_ω^0 on RHS:

$$(1 - \Phi(\bar{\omega}_j, \omega))^\eta = \frac{a' - b'e\left[X_j + Y\right] - W\gamma_\omega^0}{\kappa W}$$

And take the $1/\eta$'th root:

$$1 - \Phi(\bar{\omega}_j, \omega) = \left(\frac{a' - b'e\left[X_j + Y\right] - W\gamma_\omega^0}{\kappa W}\right)^{\frac{1}{\eta}}$$

Lastly, I subtract 1 on both sides, multiply by -1, and denote the threshold as $\Phi^*(\cdot)$ holding all terms constant:

$$\Phi^*(\bar{\omega}_j, \omega) = 1 - \left(\frac{a' - b'e\left[X_j + Y\right] - W\gamma_\omega^0}{\kappa W}\right)^{\frac{1}{\eta}}$$

Differentiating $\Phi^*(\cdot)$ with respect to each primitive gives the comparative statics reported in Section 4:

$$\frac{\partial \Phi^*(\cdot)}{\partial a'} < 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial b'} > 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial X} > 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial Y} > 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial W} > 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial \kappa} > 0, \quad \frac{\partial \Phi^*(\cdot)}{\partial \gamma_\omega^0} > 0$$

D.6 Labor demand and the role of occupational composition

Because labor is the only factor, the firm's variable labor demand is $\ell_j = \int_{\Omega_j} \gamma_j(\omega) x_j(\omega)\, d\omega$, and the framework delivers a direct link between scope and measured productivity: an increase in product range lowers measured labor productivity when outputs are aggregated

with uniform or price weights, and leaves it unchanged under marginal-cost weights, since technology itself is unaffected by the shock. Products further from the core carry higher labor requirements, so expansion mechanically dilutes the average.

Labor is homogeneous in this setup, entering through a single wage W. The measurement strategy of this paper decomposes γ into its occupational content: writing $\ell_o(\omega)$ for the requirement of occupation o per unit of ω, marginal cost becomes $c_j(\omega) = \sum_o W_o \ell_o(\omega)$, and the product-level occupational vector $\mathcal{O}(\omega)$ of Section 3.1 is the normalized profile of these requirements. Under a common wage this collapses to $W\gamma(\omega)$ and nothing is lost. The substantive content of the extension is that $\Phi(\bar{\omega}_j, \omega)$ measures the overlap in $\{\ell_o\}$ between the core and the candidate, and that overlap is what determines whether the adaptation cost $\kappa(1-\Phi)^\eta$ is paid through hiring in unfamiliar occupations or avoided by redeploying capabilities already inside the firm. Occupation-specific wages W_o would introduce a second channel, in which the cost of distance depends on which occupations a candidate requires and not only on how many are new; this is left aside here, and Φ is weighted by skill transferability rather than by relative wages.

D.7 Objects absorbed in the empirical specification

The framework is closed by imposing symmetric Cournot equilibrium with an exogenous number of firms per country, so that industry output satisfies $Y = kfX$ for f firms in each of k countries, and equilibrium scope and scale are jointly determined. I do not impose symmetry, since the empirical unit is a heterogeneous firm observed alongside its actual portfolio. Instead, the net surplus defined in Equation (6), $s_{j\omega} = a' - W\gamma^0_\omega - b'e(X_j + Y)$, collects precisely those terms that the estimating equations condition on rather than model. Demand conditions a' and b', substitutability e, industry output Y, the wage W, and the baseline requirement γ^0_ω vary at the product-year or market-year level and are absorbed by product-year fixed effects; the cannibalization term $b'eX_j$ varies at the firm-year level and is absorbed by firm-year fixed effects. What remains is variation in $\Phi(\bar{\omega}_j, \omega)$ across candidates within a firm-year, and, under the firm-candidate specification, variation within a given firm-product pair over time.

Treating the residual variation in $s_{j\omega}$ as a random draw with cdf G, as in Section 4, is

therefore a statement about the components of demand and cost that survive this conditioning. It converts the deterministic cutoff of Appendix D.5 into the adding and dropping probabilities in Equations (9) and (11), without which the model has no probabilistic content to bring to a linear probability model.

Appendix E Multi-Product Firm Analysis: Additional Results

E.1 Task-space organization by portfolio scope

Figure 6 in Section 5 pools firms of every portfolio size. Figure E1 below repeats the same exercise separately within each portfolio scope, from two to ten-or-more distinct products, to verify that the pooled relationship is not an artifact of composition.

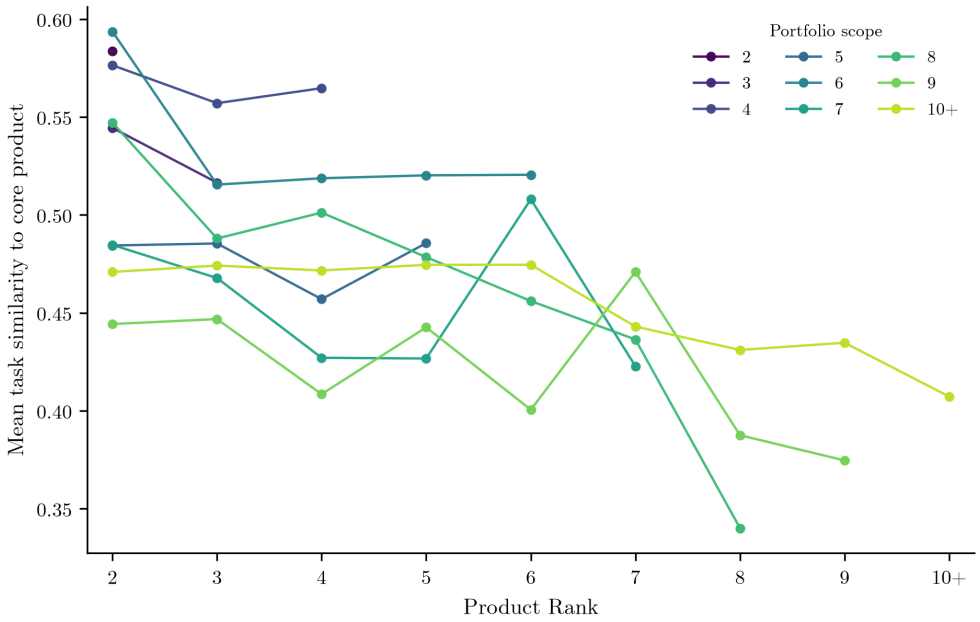


Figure E1: *This figure displays the mean task similarity (across firms and years) of co-produced products by rank, plotted separately for each portfolio scope (the number of distinct products a firm produces, top-coded at 10+). The core product (rank 1) has similarity 1 by construction in every scope and is omitted.*

E.2 Decile and percentile estimates underlying Section 5.1

This appendix reports the percentile-resolution counterpart to the decile specification of Section 5.1, together with the point estimates and standard errors underlying Figures 7, E2, 9, and E3, for the core, maximum, and portfolio similarity measures under the dynamic candidate pool.

Percentile specification. To recover the shape of the adding probability at finer resolution than the decile specification of Figure 7 allows, I estimate a percentile-based specification in which candidate products are grouped according to their similarity to the core product:

$$P(\text{Add}_{j\omega,t}) = \sum_{p=2}^{100} \beta_p \, \mathbf{1}\{\Phi(\bar{\omega}_j, \omega) \in \mathcal{Q}_p\} + \lambda_{jt} + \lambda_{\omega t} + \varepsilon_{j\omega,t}, \qquad \omega \in \mathcal{C}_{jt}^{D}, \tag{33}$$

where $\mathcal{Q}_p$ denotes the p-th percentile bin and the lowest bin is the omitted reference. Fixed effects are included to match Equation (14). Bin cutoffs are constructed over the fixed product space $\mathcal{P}$ rather than over the at-risk pool $\mathcal{C}_{jt}^{D}$, so that estimates remain comparable to the alternative static pool through sample composition alone, and not through a rescaling of the similarity distribution itself.

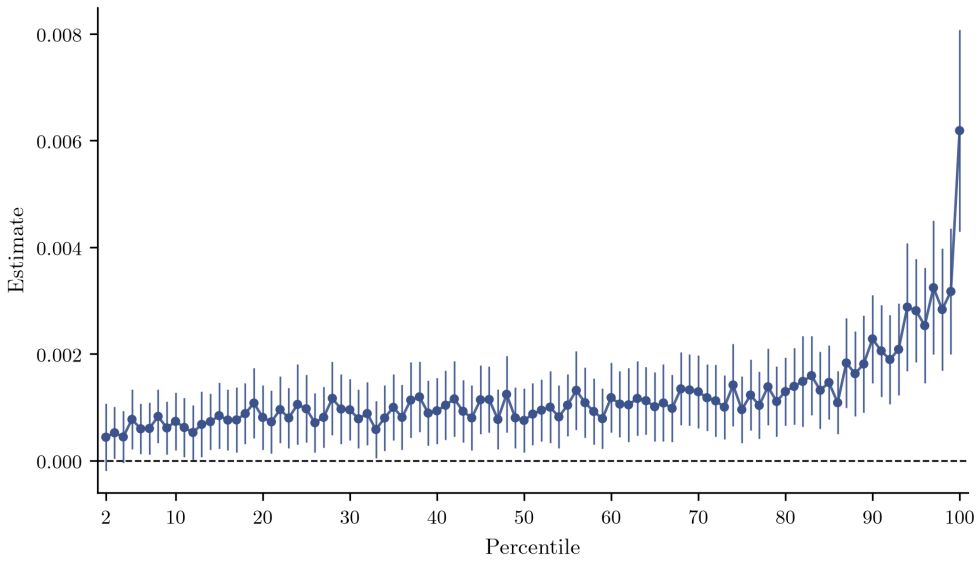


Figure E2: *This figure displays the probability of product addition by percentile of task similarity to the core product, estimated on the dynamic candidate pool.*

Figure E2 confirms and sharpens the decile pattern: the adding probability rises significantly from the 10th percentile onward, climbing smoothly through the middle of the distribution before accelerating markedly at the very top, reaching roughly 0.6 percentage points in the top percentile alone—nearly twice the 0.33 percentage point average linear effect of Table 3 and close to three times the top-decile average in Figure 7. The threshold and the curvature in Equation (9) are distinct objects, and the percentile resolution helps separate them: that the probability of addition is increasing in similarity essentially throughout the support reflects the stochastic threshold $\Phi^*_{j\omega}$ generated by demand and cost heterogeneity across firm–product pairs, whereas the sharp acceleration at the very top reflects the cost curvature $\eta < 1$.

Generalization across similarity measures. Letting $m \in \{\text{core}, \text{port}, \max\}$ index the similarity aggregator and $c \in \{D, S\}$ the candidate pool as in Equation (19), the percentile specification of Equation (33) generalizes to

$$P(\text{Add}_{j\omega,t}) = \sum_{p=2}^{100} \beta_p^{m,c} \, \mathbf{1}\{\Phi^m_{j\omega,t-1} \in \mathcal{Q}_p\} + \lambda_{jt} + \lambda_{\omega t} + \varepsilon_{j\omega,t}, \qquad \omega \in \mathcal{C}^c_{jt}. \tag{34}$$

Figure E3 below reports the percentile-level counterpart to Figure 9 in Section 5.3, at finer resolution.

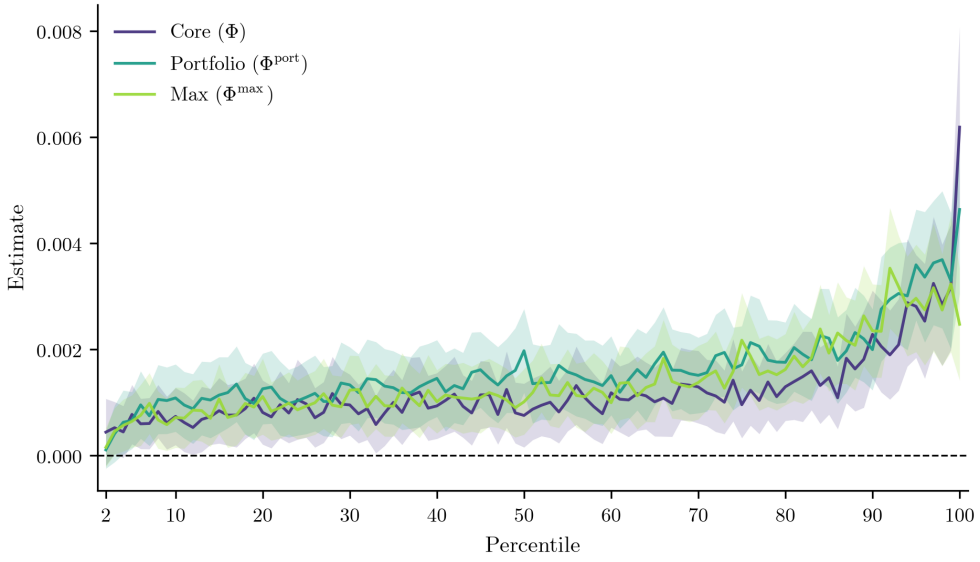


Figure E3: *This figure displays the probability of product addition by percentile of similarity to the core, maximum, and portfolio-mean measures, estimated on the dynamic candidate pool. Shaded bands are 95% confidence intervals from standard errors two-way clustered by firm and candidate product.*

Table E1: Product adding by decile of task similarity, dynamic candidate pool

	Core $\Phi(\bar{\omega}_j, \omega)$	Max $\Phi^{\max}_{j\omega,t-1}$	Portfolio $\Phi^{\text{port}}_{j\omega,t-1}$
Decile 2	0.0001	0.0002**	0.0003**
	(0.0001)	(0.0001)	(0.0001)
Decile 3	0.0003**	0.0003***	0.0003**
	(0.0001)	(0.0001)	(0.0001)
Decile 4	0.0003**	0.0004***	0.0005***
	(0.0001)	(0.0001)	(0.0001)
Decile 5	0.0004***	0.0004***	0.0007***
	(0.0001)	(0.0001)	(0.0001)
Decile 6	0.0004***	0.0006***	0.0007***
	(0.0001)	(0.0001)	(0.0002)
Decile 7	0.0006***	0.0007***	0.0008***
	(0.0001)	(0.0002)	(0.0002)
Decile 8	0.0006***	0.0010***	0.0010***
	(0.0001)	(0.0002)	(0.0002)
Decile 9	0.0009***	0.0014***	0.0012***
	(0.0002)	(0.0002)	(0.0002)
Decile 10	0.0021***	0.0022***	0.0024***
	(0.0002)	(0.0003)	(0.0003)
Fixed effects			
Firm $j \times$ Year t	Yes	Yes	Yes
Candidate product $\omega \times$ Year t	Yes	Yes	Yes
Observations	1,804,240	1,804,240	1,804,240
R^2	0.0113	0.0114	0.0114
Within R^2	0.0004	0.0005	0.0005

Notes: Sample is the dynamic candidate pool $\mathcal{C}^D_{jt}$ of Equation (13). Each column groups the corresponding similarity measure into deciles and estimates Equation (34) with 9 decile dummies; decile 1 (least similar) is the omitted reference category. Fixed effects match the baseline specification of column (3) in Tables 3–5. Standard errors, two-way clustered by firm and candidate product, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. See Figure E2 for the corresponding percentile profile.

Table E2: Product adding by selected percentiles of task similarity, dynamic candidate pool

	Core $\Phi(\bar{\omega}_j, \omega)$	Max $\Phi^{\max}_{j\omega,t-1}$	Portfolio $\Phi^{\text{port}}_{j\omega,t-1}$
Percentile 10	0.0007***	0.0007**	0.0011***
Percentile 20	0.0008***	0.0011***	0.0013***
Percentile 30	0.0010***	0.0012***	0.0013***
Percentile 40	0.0009***	0.0010***	0.0015***
Percentile 50	0.0008**	0.0010***	0.0020***
Percentile 60	0.0012***	0.0010***	0.0015***
Percentile 70	0.0013***	0.0014***	0.0015***
Percentile 80	0.0013***	0.0016***	0.0018***
Percentile 90	0.0023***	0.0023***	0.0020***
Percentile 95	0.0028***	0.0030***	0.0036***
Percentile 99	0.0032***	0.0032***	0.0033***
Percentile 100	0.0062***	0.0025***	0.0046***
Fixed effects			
Firm $j \times$ Year t	Yes	Yes	Yes
Candidate product $\omega \times$ Year t	Yes	Yes	Yes
Observations	1,804,240	1,804,240	1,804,240
R^2	0.0115	0.0115	0.0115

Notes: Sample is the dynamic candidate pool $\mathcal{C}^D_{jt}$ of Equation (13). Each column groups the corresponding similarity measure into 100 percentile bins and estimates Equation (34); percentile 1 (least similar) is the omitted reference category. For readability this table reports every tenth percentile plus the 95th, 99th and 100th; standard errors, all two-way clustered by firm and candidate product, are omitted here but the full 99-coefficient profile underlies Figure E2 (core measure) and Appendix E.2 (max and portfolio measures). Fixed effects match the baseline specification of column (3) in Tables 3–5. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

E.3 Static candidate pool: robustness

Tables E3–E5 repeat the linear adding specifications of Sections 5.1 and 5.3 under the static candidate pool $\mathcal{C}_{jt}^{S} = \mathcal{P}$, which holds the risk set fixed at the full product space in every firm–year rather than contracting it as products are added. Point estimates are close to their dynamic-pool counterparts in Tables 3, 5, and 6 throughout, and remain significant at conventional levels in every specification but the most saturated.

Figures E4 and E5 report the corresponding decile and percentile profiles for all three similarity measures, to be compared with Figures 9 and E3 under the dynamic pool; Table E6 and Table E7 report the underlying estimates.

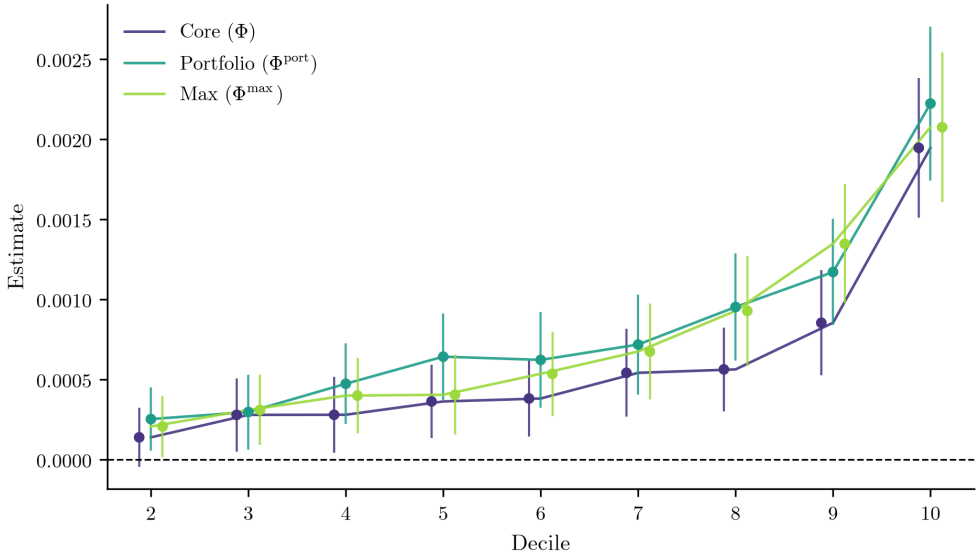


Figure E4: *This figure displays the probability of product addition by decile of similarity to the core, maximum, and portfolio-mean measures, estimated on the static candidate pool. Vertical lines are 95% confidence intervals from standard errors two-way clustered by firm and candidate product.*

Table E3: Product adding and similarity to core, static candidate pool

	(1)	(2)	(3)	(4)
	Dependent variable: $\text{Add}_{j\omega,t}$			
$\Phi(\bar{\omega}_j, \omega)$	0.0021***	0.0025***	0.0029***	0.0014***
	(0.0003)	(0.0003)	(0.0004)	(0.0005)
Fixed effects				
Firm $j \times$ Year t	No	Yes	Yes	Yes
Candidate product $\omega \times$ Year t	No	No	Yes	No
Core sector $\times$ candidate product $\omega \times$ Year t	No	No	No	Yes
Observations	1,822,135	1,822,135	1,822,135	1,822,135
R^2	0.0004	0.0082	0.0110	0.1446
Within R^2	–	0.0004	0.0004	0.0000

Notes: Sample is the static candidate pool $\mathcal{C}^S_{jt} = \mathcal{P}$, which holds the risk set fixed at the full product space in every firm–year. $\Phi(\bar{\omega}_j, \omega)$ is task similarity between the candidate product and the firm's core product, estimated as in Equation (19) with $(m, c) = (\text{core}, S)$; compare with the dynamic-pool baseline in Table 3. Column (1) reports the unconditional correlation; column (2) adds firm-by-year fixed effects; column (3) adds candidate-product-by-year fixed effects; column (4) replaces the candidate-product-by-year fixed effects with candidate-product-by-core-sector-by-year fixed effects. Standard errors, two-way clustered by firm and candidate product, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table E4: Product adding and maximum task similarity, static candidate pool

	(1)	(2)	(3)	(4)
	Dependent variable: $\text{Add}_{j\omega,t}$			
$\Phi^{\max}_{j\omega,t-1}$	0.0030***	0.0023***	0.0029***	0.0016***
	(0.0004)	(0.0003)	(0.0003)	(0.0004)
Fixed effects				
Firm $j \times$ Year t	No	Yes	Yes	Yes
Candidate product $\omega \times$ Year t	No	No	Yes	No
Core sector $\times$ candidate product $\omega \times$ Year t	No	No	No	Yes
Observations	1,822,135	1,822,135	1,822,135	1,822,135
R^2	0.0010	0.0082	0.0111	0.1446
Within R^2	–	0.0005	0.0005	0.0001

Notes: Sample is the static candidate pool $\mathcal{C}^S_{jt} = \mathcal{P}$. $\Phi^{\max}_{j\omega,t-1}$ is the leave-one-out maximum task similarity between the candidate product and the firm's lagged portfolio (Equation (17)), estimated as in Equation (19) with $(m, c) = (\max, S)$. Column (1) reports the unconditional correlation; column (2) adds firm-by-year fixed effects; column (3) adds candidate-product-by-year fixed effects; column (4) replaces the candidate-product-by-year fixed effects with candidate-product-by-core-sector-by-year fixed effects. Standard errors, two-way clustered by firm and candidate product, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table E5: Product adding and portfolio task similarity, static candidate pool

	(1)	(2)	(3)	(4)
	Dependent variable: $\text{Add}_{j\omega,t}$			
$\Phi^{\text{port}}_{j\omega,t-1}$	0.0024***[a]	0.0028***	0.0036***	0.0022***
	(0.0003)[a]	(0.0004)	(0.0004)	(0.0006)
Fixed effects				
Firm $j \times$ Year t	No	Yes	Yes	Yes
Candidate product $\omega \times$ Year t	No	No	Yes	No
Core sector $\times$ candidate product $\omega \times$ Year t	No	No	No	Yes
Observations	1,822,135	1,822,135	1,822,135	1,822,135
R^2	0.0008	0.0082	0.0111	0.1446
Within R^2	–	0.0004	0.0005	0.0001

Notes: Sample is the static candidate pool $\mathcal{C}^S_{jt} = \mathcal{P}$. $\Phi^{\text{port}}_{j\omega,t-1}$ is the leave-one-out, sales-share-weighted mean task similarity between the candidate product and the firm's lagged portfolio (Equation (16)), estimated as in Equation (19) with $(m, c) = (\text{port}, S)$. Column (1) reports the unconditional correlation; column (2) adds firm-by-year fixed effects; column (3) adds candidate-product-by-year fixed effects; column (4) replaces the candidate-product-by-year fixed effects with candidate-product-by-core-sector-by-year fixed effects. Standard errors, two-way clustered by firm and candidate product except where noted, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

[a] The two-way-clustered export for this cell was corrupted (the underlying workbook's raw-correlation sheet duplicated the firm-by-year specification); the coefficient and firm-clustered standard error shown are taken from the companion single-way-clustered export instead. Re-export this cell with two-way clustering when the source file is regenerated.

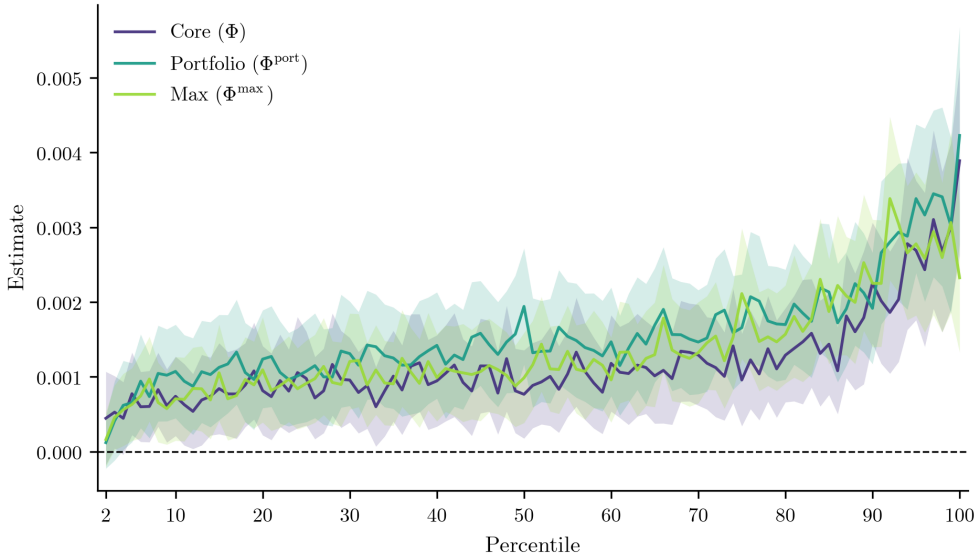


Figure E5: *This figure displays the probability of product addition by percentile of similarity to the core, maximum, and portfolio-mean measures, estimated on the static candidate pool. Shaded bands are 95% confidence intervals from standard errors two-way clustered by firm and candidate product.*

Table E6: Product adding by decile of task similarity, static candidate pool

	Core $\Phi(\bar{\omega}_j, \omega)$	Max $\Phi^{\max}_{j\omega,t-1}$	Portfolio $\Phi^{\text{port}}_{j\omega,t-1}$
Decile 2	0.0001	0.0002**	0.0003**
	(0.0001)	(0.0001)	(0.0001)
Decile 3	0.0003**	0.0003***	0.0003**
	(0.0001)	(0.0001)	(0.0001)
Decile 4	0.0003**	0.0004***	0.0005***
	(0.0001)	(0.0001)	(0.0001)
Decile 5	0.0004***	0.0004***	0.0006***
	(0.0001)	(0.0001)	(0.0001)
Decile 6	0.0004***	0.0005***	0.0006***
	(0.0001)	(0.0001)	(0.0002)
Decile 7	0.0005***	0.0007***	0.0007***
	(0.0001)	(0.0002)	(0.0002)
Decile 8	0.0006***	0.0009***	0.0010***
	(0.0001)	(0.0002)	(0.0002)
Decile 9	0.0009***	0.0013***	0.0012***
	(0.0002)	(0.0002)	(0.0002)
Decile 10	0.0019***	0.0021***	0.0022***
	(0.0002)	(0.0002)	(0.0002)
Fixed effects			
Firm $j \times$ Year t	Yes	Yes	Yes
Candidate product $\omega \times$ Year t	Yes	Yes	Yes
Observations	1,822,135	1,822,135	1,822,135
R^2	0.0110	0.0111	0.0110
Within R^2	0.0004	0.0005	0.0004

Notes: Sample is the static candidate pool $\mathcal{C}^S_{jt} = \mathcal{P}$. Each column groups the corresponding similarity measure into deciles and estimates Equation (34) with 9 decile dummies; decile 1 (least similar) is the omitted reference category. Fixed effects match the baseline specification of column (3) in Tables E3–E5. Standard errors, two-way clustered by firm and candidate product, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table E7: Product adding by selected percentiles of task similarity, static candidate pool

	Core $\Phi(\bar{\omega}_j, \omega)$	Max $\Phi^{\max}_{j\omega,t-1}$	Portfolio $\Phi^{\text{port}}_{j\omega,t-1}$
Percentile 10	0.0007***	0.0007**	0.0011***
Percentile 20	0.0008***	0.0011***	0.0012***
Percentile 30	0.0010***	0.0012***	0.0013***
Percentile 40	0.0009***	0.0010***	0.0014***
Percentile 50	0.0008**	0.0010***	0.0019***
Percentile 60	0.0012***	0.0010***	0.0015***
Percentile 70	0.0013***	0.0013***	0.0015***
Percentile 80	0.0013***	0.0016***	0.0017***
Percentile 90	0.0023***	0.0022***	0.0019***
Percentile 95	0.0027***	0.0028***	0.0034***
Percentile 99	0.0030***	0.0031***	0.0030***
Percentile 100	0.0039***	0.0023***	0.0042***
Fixed effects			
Firm $j \times$ Year t	Yes	Yes	Yes
Candidate product $\omega \times$ Year t	Yes	Yes	Yes
Observations	1,822,135	1,822,135	1,822,135
R^2	0.0111	0.0111	0.0111

Notes: Sample is the static candidate pool $\mathcal{C}^S_{jt} = \mathcal{P}$. Each column groups the corresponding similarity measure into 100 percentile bins and estimates Equation (34); percentile 1 (least similar) is the omitted reference category. For readability this table reports every tenth percentile plus the 95th, 99th and 100th; standard errors, all two-way clustered by firm and candidate product, are omitted here but the full 99-coefficient profile is available in the underlying export. Fixed effects match the baseline specification of column (3) in Tables E3–E5. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

E.4 Product dropping by portfolio rank

Table E8 reports the point estimates and standard errors underlying Figure 8 in Section 5.2.

Table E8: Product dropping by sales rank within the portfolio

	Dependent variable: $\text{Drop}_{j\omega,t+1}$
Rank 2	0.0048
	(0.0034)
Rank 3	0.0383***
	(0.0066)
Rank 4	0.0765***
	(0.0097)
Rank 5	0.0853***
	(0.0117)
Rank 6	0.1179***
	(0.0155)
Rank 7	0.1392***
	(0.0182)
Rank 8	0.1532***
	(0.0247)
Rank 9	0.1570***
	(0.0259)
Rank 10+	0.2676***
	(0.0190)
Fixed effects	
Firm $j \times$ Year t	Yes
Incumbent product $\omega \times$ Year t	Yes
Observations	14,824
R^2	0.4845
Within R^2	0.0681

Notes: Sample is the firm's incumbent portfolio Ω_{jt}, grouped by sales rank within the portfolio in period t; rank 1 (the top-selling, and typically the core, product) is the omitted reference category. Fixed effects match column (3) of Table 4. Standard errors, two-way clustered by firm and product, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

E.5 Product dropping excluding the core product

Table E9 repeats the linear dropping specification of Section 5.2, excluding the firm's core product from the incumbent sample. The core product is dropped with near-zero frequency and would otherwise be a mechanical outlier; excluding it, the coefficient on task similarity retains its sign but loses precision as fixed effects are progressively saturated, reflecting the much smaller and noisier sample of non-core incumbent products.

Table E9: Product dropping and task similarity to the core product, excluding the core product

	(1)	(2)	(3)	(4)
	Dependent variable: $\text{Drop}_{j\omega,t+1}$			
$\Phi(\bar{\omega}_j, \omega)$	-0.0562***	-0.0509***	-0.0297	-0.0437
	(0.0125)	(0.0157)	(0.0182)	(0.0444)
Fixed effects				
Firm $j \times$ Year t	No	Yes	Yes	Yes
Incumbent product $\omega \times$ Year t	No	No	Yes	No
Core sector $\times$ incumbent product $\omega \times$ Year t	No	No	No	Yes
Observations	11,300	11,300	11,300	11,300
R^2	0.0042	0.2843	0.5157	0.8407
Within R^2	–	0.0020	0.0005	0.0004

Notes: Same sample and specification as Table 4, restricted to incumbent products other than the firm's core product itself (the core product is dropped with near-zero frequency and would otherwise be a mechanical outlier). The coefficient loses precision and significance once product-by-year fixed effects are added, indicating that the dropping result in Table 4 is not driven solely by the core product's own high similarity and low dropping probability, but the smaller, noisier sample here is less informative in the more saturated specifications. Standard errors, two-way clustered by firm and product, are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.